
SPATIOTEMPORALLY LEARNING THE MORPHOLOGICAL AND DYNAMIC FEATURES OF AU MELTPOOL UNDER DIFFERENT HEAT SOURCE TYPES WITH PHASE-FIELD INFORMED U-NET FRAMEWORK

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ABSTRACT

The uncertainties in the interaction between laser and microstructural interfaces pose a major hindrance to precision engineering in additive manufacturing. Proper assessment of the role of laser heat source types on microstructure evolution and phase transitions is crucial for understanding the process-property relationship of 3D-printed parts. To address this, we present a multiphysics computational study that explores thermal distribution, phase transformation, and melt-pool flow across different laser beam profiles and free energy expressions in a unary Au system. A detailed comparison of the microstructural pathways developed under four laser beam shapes (Gaussian, Flat-Top, Ring, and Bessel) was conducted, along with two different mathematical expressions for free energy. Furthermore, a two-stage U-Net architecture was implemented using the simulation data. The initial network predicted phase from temperature inputs, achieving accuracies of 0.9992, 0.9989, and 0.9979 for training, validation and testing sets, respectively. The second stage employed temperature and phase to predict velocity, yielding MSE of 0.068, MAE of 0.079, and R^2 of 0.9990 on test data. For user-supplied temperature distribution array, the built primary MicroTwin can predict phase evolution, whereas a secondary FlowTwin informed by MicroTwin's prediction is capable of estimating velocity selectively inside the liquid meltpool domain.

Keywords · Moving Heat Source · Super Gaussian Order · FCC Au · MicroTwin · Phase-informed FlowTwin · Conditional Automation

1 Introduction

The advancement of intelligent manufacturing has transformed conventional process design through the integration of computational modelling, machine learning, and real-time data analysis to achieve adaptive control and quantify uncertainties in complex fabrication processes [1]. In laser-aided Additive Manufacturing (AM), uncertainties in meltpool geometry, area, and morphology directly influence the resulting microstructure and mechanical performance of the fabricated component [2, 3]. Addressing these uncertainties requires a detailed understanding of the coupled thermo-physical phenomena governing meltpool evolution under varying processing conditions. Recent progress in multi-physics simulations and data-driven approaches now enables the study of these phenomena across multiple spatio-temporal scales, paving the way for intelligent systems capable of predicting and controlling microstructural evolution in real-time [4]. The integration of unified phase-field modeling within an ICME (Integrated Computational Materials Engineering) framework has proven instrumental in accurately predicting microstructural evolution [5] and optimizing process parameters for high-performance alloys treated by lasers of varying heat source types.

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Building, upon this motivation, this study presents a comprehensive computational analysis of how four different laser beam profiles affect the microstructure evolution in a unary Au-system during laser processing. Using Phase-Field (PF) simulations, we traced the melt pool geometry, its area, and the thermal history within the computational domain. A quantitative comparison between two different free energy expressions driving the phase transition was also carried out. The first approach used a parabolic free energy function fitted to thermodynamic data (fine-tuned expression), while the second relied on case equations extracted directly from a Thermodynamic Database (TDB) file using a Python script. Comparisons of free energy expressions in phase-field modeling are rare, as exact thermodynamic expressions are difficult to obtain, leading most studies to rely on polynomial fits [6]. Moreover, many such studies are often limited to free energy expressions defined at two fixed temperatures for two phases [7]. In this work, we have integrated thermodynamic expressions for free energies as a function of temperature, making the phase transition process inherently non-isothermal. Our study employs the MOOSE Framework as the multi-physics simulation environment [8, 9], where three coupled physical phenomena are solved: heat-transfer, phase-field and fluid-flow, using Finite Element Method (FEM).

As described in our previous work [10], most computational studies of laser processing treats thermal and phase diffusion as identical. This assumption leads to the presumption that the LIQUID phase (melt pool) forms instantaneously upon reaching melting temperature, overlooking the real physical behavior in laser-based additive manufacturing where phase diffusion lags behind the thermal diffusion front [11]. In this study, we found that implementing the TDB-based free energy expression replicates the real-world behavior of the lead-lag dynamics between phase and thermal fronts. The main reason this distinction is often overlooked in contemporary research is likely computational cost and efficiency. However, ignoring it fails to capture the intricate interplay between the two diffusion fronts, which in turn influences other physical phenomena such as fluid flow, thermal conductivity, and phase densities. In addition to phase-field modeling, the FEM simulations generated data were used to train machine learning models capable of predicting the LIQUID phase, its shape, areas, and its velocity, using the temperature distribution of the domain as input.

This approach provides a foundation for realizing the conditional automation in additive manufacturing. Conditional automation refers to systems that adaptively control manufacturing processes based on real-time sensor inputs or evolving process conditions. Unlike fixed-sequence automation, these systems respond dynamically to parameters such as temperature, melt pool shape, or phase evolution, improving precision and consistency through live feedback control algorithms. Phase-field simulations and U-Net based architectures form the core building blocks for achieving such adaptive, data-driven process control conditional automation of additive manufacturing [12, 13].

In the following section, we discuss in detail, the types of laser heat sources, the expressions for thermodynamic driving forces, temperature-dependent material properties, and machine learning architectures. We also outline the conceptualization of physics informed virtual digital-twin for laser processing using the trained ML model. Finally, in the Results and Discussion (Section 8), we explore the effects on phase evolution, temperature distribution, and melt pool flow characteristics under different laser heat sources, and distinct free-energy expressions. For comparison of the laser beam profiles, the laser power (P), scan speed (v), and beam outer radius (r_x) were kept constant across all simulations.

2 Initial and Boundary Conditions

Simulations were carried out in a computational domain of $1000 \mu m \times 250 \mu m$ in the x-z plane, as shown in Figure 1. The system consists of two phases: LIQUID representing molten gold, and FCC representing solid gold. At the initial time (t_0), a laser beam with power of $25 mW$ was irradiated on the top surface at a distance of $125 \mu m$ from the left, with a scan speed of $30 \mu m/s$. The initial area of melt pool was designed to be $2500 \mu m^2$. The computational domain is surrounded by an ambient environment at a temperature $T_{air} = 300 K$ on three sides (left, right, and top), while the bottom rests on an interface with a heat transfer coefficient $h_{int} = 11500 W/m^2 K$ [14]. The convective heat transfer coefficient of the surrounding air is taken as $h_{amb} = 50 W/m^2 K$ [15], while the emissivity for radiative heat transfer is set to 0.3 [16]. Similarly the melt pool velocity field is initialized to zero everywhere ($\vec{u}_{t_0} = \vec{0}$), reflecting a quiescent state prior to laser irradiation. The initial LIQUID phase starts at uniform temperature, inducing no initial gradients. Once the laser heat deposition (Q_{laser}) starts, it rapidly generates ∇T , which via temperature-dependent properties, triggers buoyancy and dominant Marangoni convection and the FEM solver evolves $\vec{u}$ self-consistently from this rest state, capturing ramp-up transients without spurious artifacts. To accurately assign the material properties of different phases, including densities, specific heat capacity, thermal conductivities, and melt pool viscosity, under non-isothermal conditions, these properties were defined as the function of temperature. The detailed formulations are provided in Section 6.

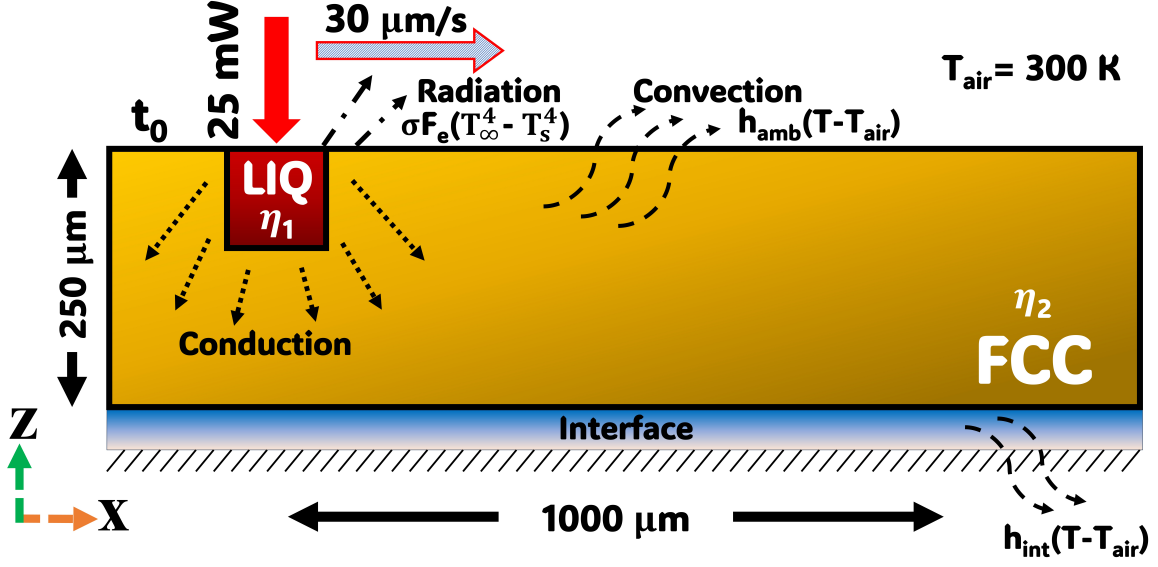


Figure 1: Schematic of the computational domain setup showing laser-induced phase evolution in the Phase-Field Model at time t_0 . The domain represents a $1000 \mu\text{m} \times 250 \mu\text{m}$ section of a unary Au system in the x-z plane. The system rests on a fixed interface at the bottom, with ambient air on the remaining three sides. Heat enters the domain by conduction and exits through radiation and convection at the air and interface boundaries. A laser heat source of 25 mW irradiates the top surface positioned $125 \mu\text{m}$ from the top-left corner, moving at a scan speed of $30 \mu\text{m/s}$ (indicated by the red arrow above the LIQUID phase). The initial LIQUID phase region (η_1 , $2500 \mu\text{m}^2$) represents the melt pool shown in red, while the remaining solid region is the FCC phase (η_2 shown in yellow). The same computational setup, phase structure, and processing parameters (laser power and scanspeed) are maintained across all the simulations with different beam profiles.

3 Non-isothermal phase field model

The phase field model consists of two phases: LIQUID (melt pool) and FCC (solid), which are represented by the non-conserved order parameter ' η ' and ' $1 - \eta$ ' respectively. The value of ' η ' is set to 1 inside the LIQUID phase and 0 at the FCC phase. The conserved variable 'composition' remains constant throughout the domain as this work follows the evolution of phases in a single-component system. Thus, this study focuses on the two phases and their transformation induced by laser heat deposition. For this setup, a swift transition of material properties between phases with thermodynamic consistency is required which is achieved using the interpolation function $h_i(\eta)$ introduced by Moelans [17], as presented in Equation 1.

$$h_i = \frac{\eta_i^2}{\sum_{k=1}^N \eta_k^2} \quad (1)$$

In the above equation, N and i represents total number of order parameters and specific phase in the system. The interpolation function h_i ensures a thermodynamically consistent transition in the mushy zone, where intermediate values of η ($0 < \eta < 1$) correspond to mixed solid-liquid regions with phase fractions h_{FCC} and h_{LIQUID} , facilitating smooth property blending without unphysical discontinuities. This diffuse interface representation, with an effective mushy zone thickness on the order of $\delta = 10 \mu\text{m}$, aligns with experimental observations of solidification microstructures in alloys [18]. The driving force dictating the morphological changes of microstructure in the domain is the total free energy of the system. With an external heat source, the total free energy (F_{total}) can be modeled as the combination of bulk free energy (F_{bulk}) and interfacial free energy (F_{int}), as given by Equation 2.

$$F_{total} = F_{bulk} + F_{int} = \int_V (f_{bulk} + f_{int}) dV \quad (2)$$

here, the bulk and interfacial energies are expressed in terms of free energy densities (energy per unit molar volume of Au). The bulk free energy density (f_{bulk}) is expressed in Equation 3.

$$f_{bulk} = \sum_i h_i f_{bulk}^i = \sum_i h_i \frac{F_i(T)}{V_m} \quad (3)$$

In Equation 3, h_i represents the interpolation function, whereas f_{bulk}^i denotes the bulk free energy density of the i^{th} phase. V_m is the molar volume of gold, with a value of $10.21 \text{ cm}^3/\text{mol}$ [19], taken for both phases. Additionally, $F_i(T)$ represents the temperature-dependent Gibbs free energy of phase ' i '. The correct formulation of the Gibbs free energy is necessary for accurate description of the phase and melt pool evolution during laser processing of metallic microstructures. In the Section 4, two variants of bulk free energy expressions: TDB-based ($F_{bulk} = F_\psi$) and fine-tuned ($F_{bulk} = F_\theta$) are formulated, for assessment of the quantitative aspects of the driving force in the evolution model. The latent heat L_f represents the enthalpy absorbed/released during the solid-liquid phase change [20] and is thermodynamically encoded in the free energy expressions via the entropy discontinuity $\Delta S = L_f/T_m$ at T_m . The expression for L_f , can be postprocessed in the phase field model as given by the following equation: $L_f = \frac{d\Delta F}{dT} T_m$, where ΔF is difference in gibbs free energy of FCC and LIQUID phases at the vicinity of the melting point and is provided in Figure A1 of Appendix A. A steeper $|d\Delta F/dT|$ implies a sharper ramp-up of $|\Delta F|$ above T_m , confining the effective transition to a narrow temperature window. In the phase-field model, L_f is thus implicitly incorporated through the temperature-dependent bulk free energy density $f_{bulk} = \sum_i h_i F_i(T)/V_m$ (Equation 3), which drives the order parameter evolution and ensures the integrated driving force over the full phase transition equals L_f per mole.

Similarly, the interfacial free energy density (f_{int}) in Equation 2 is further expressed in Equation 4.

$$f_{int} = m f_{barrier} + \frac{\kappa}{2} \sum_i (\nabla \eta_i)^2 \quad (4)$$

where, m and κ are the model parameters associated with interfacial energy (σ [J/m^2]) and diffused interface width (δ [m]) as follows:

$$m = \frac{6\sigma}{\delta} \quad (5)$$

$$\kappa = \frac{3}{4} \delta \sigma \quad (6)$$

In this study, we have set the value of the interface width δ to be $10 \mu m$ and the interfacial energy to 0.5 J/m^2 . The term $f_{barrier}$ in Equation 4 is a fourth-order Landau polynomial of the phase field variable, as expressed in Equation 7 with the value of γ taken as 1.5 [21].

$$f_{barrier} = \sum_i \left(\frac{(\eta_i)^4}{4} - \frac{(\eta_i)^2}{2} \right) + \gamma \sum_i \sum_{j \neq i} \eta_i^2 \eta_j^2 + \frac{1}{4} \quad (7)$$

During laser processing, the spatio-temporal tracking of non-conserved phase field variable i.e., the order parameter (η), provides an accurate description of phase transformation. Equation 8 describes such evolution of this variable as:

$$\frac{\partial \eta_i}{\partial t} = -L(\eta_1, \dots, \eta_N) \left(\frac{\partial f_{bulk}}{\partial \eta_i} - \nabla \cdot \frac{\partial f_{int}}{\partial \nabla \eta_i} \right) \quad (8)$$

where, $L(\eta_1, \dots, \eta_N) = L^i$ which is defined as follows:

$$L^i = \frac{\sum_{i=1}^N \sum_{j>1}^N L_{ij} \eta_i^2 \eta_j^2}{\sum_{i=1}^N \sum_{j>1}^N \eta_i^2 \eta_j^2} \quad (9)$$

The interfacial mobility in Equation 9 is further approximated as follows:

$$L_{ij} = \frac{4m}{3\kappa} \times \frac{M^i + M^j}{2} \quad (10)$$

here, M^i and M^j are the phase mobilities of the two adjacent phases. Similarly, for the heat deposited by the laser source, heat transfer follows the partial differential equation presented in Equation 11.

$$\rho^i C_p^i \left(\frac{\partial T}{\partial t} + \vec{u} \cdot \nabla T \right) = k_{th}^i \nabla^2 T + Q_{laser} \quad (11)$$

In the above equation, temperature is denoted by T at time t . The density, specific heat capacity, and thermal conductivity of phase ' i ' are represented by ρ^i , C_p^i , and k_{th}^i , respectively while the external laser heat source is denoted by Q_{laser} . The flow velocity inside the meltpool is given by $\vec{u}$ solved via Navier-Stokes Equations 12 [22] assuming the flow as incompressible, which contributes to convective heat transfer. The viscosity, surface tension, and associated Marangoni effect inside the inside meltpool flow is further discussed in Section 6.

$$\begin{aligned} \nabla \cdot \vec{u} &= 0 \\ \rho^{LIQ} \left[\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right] + \nabla \cdot \sigma - \vec{f} &= 0 \\ \sigma &= -p\mathbf{I} + \mu \left[\nabla \vec{u} + (\nabla \vec{u})^T \right] \end{aligned} \quad (12)$$

where, σ is total stress tensor, $\vec{f}$ includes the boussinesq and gravity force terms, and μ is the dynamic viscosity.

4 Thermodynamic Driving Force Formulations for Heat-treated Metallic Gold

For computational efficiency and faster numerical convergence, polynomial expressions of free energy are often preferred, especially parabolic expressions of second-order polynomials, which are widely used for modeling binary and ternary alloy systems [23, 24, 25]. Although, polynomial expressions with $n>2$ generally offer more accurate descriptions, they are computationally expensive due to the compositional complexities they introduce into the diffusion equations. In this study on laser processing of pure element, the free energy depends solely on temperature (T), with constant composition and pressure. This condition allows for the flexibility to expand the polynomial expression beyond traditional parabolic forms to include logarithmic terms. Accordingly, the free energy for pure Au under temperature change is best described by the fine-tuned expression (F_θ), as presented in Equation 13, derived in a manner similarly to [26, 27]. Similarly, the TDB-extracted expressions (F_ψ) are piecewise continuous as presented in Equation 13. These expressions are based on Thermodynamic Data Base file formulated by Khvan et al. [28]. In Equation 13, the coefficients $A, B, C, D, E, F, G, H, I, J$ and T_{th} , with subscripts ' θ ' and ' ψ ', represent the thermodynamic coefficients of the free energy function for the fine-tuned and TDB-based expressions, respectively. The coefficient values are listed in Table 1 for both the LIQUID and FCC phases. The variable ' T ' depicts the temperature dependence of the free energy in both expressions.

The two free-energy formulations were combined with four different laser beam profiles to analyze their effects on phase growth, while keeping other laser parameters constant. The expressions, and their temperature-dependent plots are presented in Equation 13 and Figure 2(a) and (b), respectively. As shown in Figure 2, the F_θ curves for the FCC and LIQUID phases display a distinct phase-change interval, while the F_ψ curves derived from the TDB expressions do not exhibit this feature.

$$F_\theta = T_{th} \left[A_\theta - B_\theta - C_\theta(T-T_0) + D_\theta(T-T_0) \ln(E_\theta T) + F_\theta(T-T_0)^2 + G_\theta(T-T_0)^3 + H_\theta T^{-1} \right]$$

$$F_\psi = \begin{cases} LIQUID & : A_\psi T \ln \left[1 - \exp(-B_\psi T^{-1}) \right] - RT \ln \left[1 + \exp \left(C_\psi (D_\psi - E_\psi T) T^{-1} \right) \right] - F_\psi T^2 - G_\psi \\ FCC & : A_\psi T \ln \left(1 - \exp(-B_\psi T^{-1}) \right) - C_\psi + \begin{cases} T < 1337.33 & : D_\psi T^2 - E_\psi T^3 \\ T \geq 1337.33 & : F_\psi T^{-11} - G_\psi T^{-5} - H_\psi T \ln(T) + I_\psi T + J_\psi \end{cases} \end{cases} \quad (13)$$

Table 1: Coefficients of thermodynamic constants for the Free energy expressions.

Coeff.	LIQUID	FCC	Unit	Coeff.	LIQUID	Unit	FCC	Unit
A_θ	1337	300	J/mol	A_ψ	24.9435	J/mol K	24.9435	J/mol K
B_θ	24474	515	J/mol	B_ψ	126.6874	K	126.6874	K
C_θ	16.2	-46.2	J/mol K	C_ψ	-0.12027		6103.6566	J/mol
D_θ	24.5	-13.5	J/mol K	D_ψ	19700	K	-0.001281	J/mol K ²
E_θ	0.00214	0.1985	1/K	E_ψ	14.917		-4.02×10^{-7}	J/mol K ³
F_θ	0.04512	-0.001	J/mol K ²	F_ψ	0.00067	J/mol K ²	1.55×10^{36}	JK ¹¹ /mol
G_θ	1.9×10^{-5}	2.0×10^{-5}	J/mol K ³	G_ψ	326.3861	J/mol	-2.27×10^{18}	JK ⁵ /mol
H_θ	-17.4	-0.001	JK/mol	H_ψ			-2.0566	J/mol K
T_0	2662	121	K	I_ψ			9.8072	J/mol K
T_{th}	10.412	9.03		J_ψ			3898.778	J/mol

A sharp or narrow phase-change interval, such as in pure metals, causes rapid phase transition during melting and solidification. This can result in residual stresses [29, 30], and may complicate numerical convergence [31]. In contrast, when modeling compositionally-complex alloys, a broader phase-change interval can play an important role in ensuring stability and convergence. Thus, a fine-tuned (fitted) free energy function can be effectively applied to multicomponent system, even with only limited energy-temperature thermodynamic data are available. On the other hand, when a thermodynamic database (.TDB) file is available, free energy expressions can be extracted directly and implemented into a FEM solver using a simple Python script, thereby formulating the required temperature-dependent expressions. This direct TDB-based approach not only ensures thermodynamic consistency but also enhances the accuracy. As observed in Figure 2(b), the near absence of a phase-change interval implies a very narrow temporal range over which multiple phases co-exist. This leads to rapid interpolation of material and phase-dependent properties, thereby improving the numerical stability of the model.

5 Laser Beam Heat Source Models

The Gaussian laser beam is the most widely used profile for modeling laser-based melting and solidification in additive manufacturing, owing to its simple mathematical form. The beam peaks intensity at the center and decays outwards exponentially, creating steep thermal gradients that influences microstructure evolution [32, 33, 34]. However, uniform heating is often desired in practice to reduce thermal gradients, motivating alternative beam profiles such as the Flat-Top beam, which truncates the Gaussian peak to produce a uniform intensity region, and the Ring-shaped beam, which distributes heat in an annular pattern with maximum intensity at a specific radius [35]. The Bessel beam combines both characteristics, featuring a Gaussian-like center surrounded by concentric rings defined by the zeroth-order Bessel function [36]. Experimental studies confirm that beam shape strongly affects meltpool geometry and microstructure: Galbusera et al. [37] showed that varying beam shape alters meltpool aspect ratio and depth, while Wischeropp et al. [38] found that donut-shaped beams reduce defects in AlSi10Mg. Tumkur et al. [39] further demonstrated that Bessel beam improve control over meltpool dynamics, lowering porosity and keyhole defects. These findings highlight the critical role of beam profile in determining microstructural and mechanical properties, underscoring the value of computational modeling for deeper insights into phase transformation, heat flow and defect prediction, areas difficult to probe experimentally. In this work, we compare four laser beam profiles: Gaussian, Flat-Top, Ring and Bessel, under identical process parameters in laser-aided phase-field simulations of a pure Au system. The effects on meltpool shape, area, flow and temperature distribution are analyzed, revealing a lead-lag dynamic between heat and mass diffusion fronts.

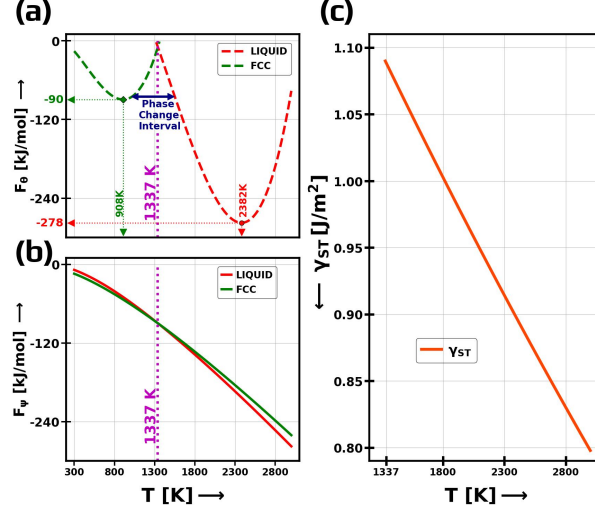


Figure 2: Graphical representation of the two free energy expressions used for phase-field simulations of laser-processed unary Au. In (a), the free energy curves are based on a higher-order polynomial-logarithmic function obtained from thermodynamic data fitting, referred to as the fine-tuned expression (F_0 , Equation 13). The dashed green and red curves corresponds to the FCC and LIQUID phases, respectively. Below the melting temperature (1337 K), the FCC phase is more stable, while above 1337 K, the LIQUID phase becomes stable. In (b), the free energy expression extracted directly from the *TDB* file, referred to as the TDB-based expression (F_ψ , Equation 13), is shown. Similar to (a), the red curve represents the LIQUID phase and the green curve the FCC phase, but here the two curves intersect at 1337 K without a distinct phase change interval. The presence or absence of this interval in (a) and (b) governs how thermal history influences structural evolution, leading to distinct simulation outcomes. (c) Variation of surface tension in the melt pool induced by the dynamic heat source, leading to thermo-capillary convection.

5.1 Gaussian Beam

Among the four heat source types, the Gaussian beam serves as the reference case due to its simple formulation and well-understood thermal behavior [40, 41, 42]. It produces maximum irradiance at the center, decaying radially to form the characteristic Gaussian distribution. The irradiance (I_G is defined in Equation 14 [36, 43], where r_G denotes the effective beam radius corresponding to $\frac{1}{e^2}$ of the peak intensity, P is the laser power, η_e is efficiency of laser power to heat conversion, and C_a and C_b are constants whose values are listed in Table 2. A representative Gaussian heat source is illustrated in Figure 3 with a 3D and cross-sectional view.

$$I_G = \frac{P}{\pi r_G^2} C_a \eta_e \exp \left[-C_b \left(\frac{r^2}{r_G^2} \right) \right] \quad (14)$$

5.2 Flat Top | Super Gaussian Beam

The Flat-Top or Super-Gaussian beam provides a near-uniform intensity across its radius with a sharp drop at the edges, enabling more homogeneous heating of the material [43, 44]. Its irradiance (I_{FT}) is defined in Equation 15 [36, 45, 43, 46], where r_{FT} is the effective beam radius. All other processing parameters P , η_e , C_a and C_b are kept identical for all four laser profiles, while the additional parameters k and $\Gamma(1/k)$ represents the super-Gaussian order, set to 4.2 in this study and gamma function respectively [36]. The Flat-Top beam profile is illustrated in the second column of Figure 3, with 3D, top and side views.

$$I_{FT} = \frac{P}{\pi r_{FT}^2} \frac{k}{\Gamma(1/k)} C_a^{1/k} \eta_e \exp \left[-C_b \left(\frac{r^2}{r_{FT}^2} \right)^k \right] \quad (15)$$

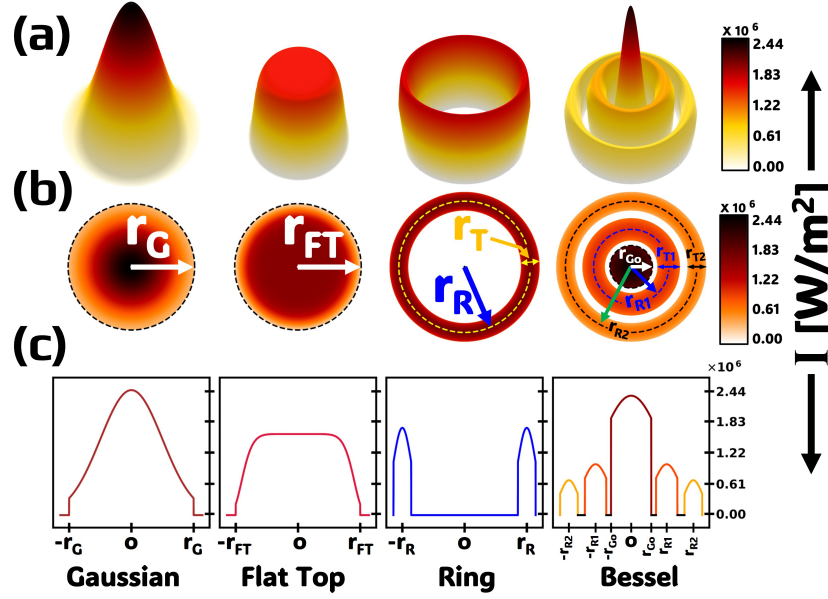


Figure 3: Visualization of four laser beam profiles- Gaussian, Flat-Top, Ring, and Bessel, used for laser-aided phase-field simulations of unary Au. (a) 3D intensity maps of the four beam types, each with an equivalent outer radius of $70\mu\text{m}$. The peak intensities at the beam center are $2.44 \times 10^6 \text{ W/m}^2$ for Gaussian, $1.58 \times 10^6 \text{ W/m}^2$ for Flat-Top, $1.70 \times 10^6 \text{ W/m}^2$ for Ring, and $2.33 \times 10^6 \text{ W/m}^2$ for Bessel beam, as shown by the color scale. (b) Top-view intensity distributions for the same beams, where r_G and r_{FT} denotes the Gaussian and Flat-Top beam radii, respectively, and r_R and r_T indicates the Ring beam radius and thickness. For the Bessel beam with a central core and two outer rings, the core radius is r_{G0} , while the outer rings have radii r_{R1} and r_{R2} with respective thickness r_{T1} and r_{T2} . (c) Side-view irradiance profile showing the variation of intensity with radial distance, where the y-axis represents irradiance and the x-axis represents the distance from the beam center.

5.3 Ring Beam

The ring laser beam features a hollow circular profile with maximum intensity along its circumference and minimal intensity at the center. This configuration promotes outward-to-inward heat flow, helping achieve more uniform temperature distribution and reduced thermal defects [47, 48]. The beam introduces one additional process parameter, the ring width, which controls the energy spread across the domain [49, 50]. The irradiance (I_R) for the ring beam is defined in Equation 16 [34, 35, 51, 52]. All parameters (P , η_e , C_a , C_b) are identical to those used in Gaussian and Flat-Top sources, while r_R and r_T represents the ring radius and ring thickness respectively.

$$I_R = \frac{P}{\pi r_T^2 Y(r_R, r_T)} C_a \eta_e \exp \left[-C_b \left(\frac{(r - r_R)^2}{r_T^2} \right) \right] \quad (16)$$

The radial distribution function $Y(r_R, r_T)$, as described by Moore et al. [34], accounts for the combined influence of ring radius and thickness through an exponential decay and complementary error function expressed as:

$$Y(r_R, r_T) = \exp \left(\frac{-r_R^2}{2r_T^2} \right) + \frac{r_R}{r_T} \sqrt{\frac{\pi}{2}} \operatorname{erfc} \left(\frac{-r_R}{\sqrt{2} r_T} \right)$$

A visual representation of the ring beam is shown in the third column of Figure 3, including 3D, top and side views in (a), (b) and (c) respectively.

5.4 Bessel Beam

The Bessel beam consists of a central core surrounded by concentric ring beams, with irradiance gradually decreasing radially outward. In this study, a central Gaussian-type beam with two concentric rings are considered, where each radius corresponds to the zeros of the zeroth-order Bessel function. While some studies define the core radius as the point where irradiance drops to $\frac{1}{e^2}$ of its peak [53], here it is taken at the first zero of the Bessel function, with the

subsequent rings located at the second and third zeros. To ensure comparable processing conditions across all beam types, the laser power is divide among the three components of the Bessel beam using attenuation coefficients a_0 , a_1 , and a_2 which satisfy as $a_0 + a_1 + a_2 = 1$. The irradiance (I_B) is expressed in Equation 17 [36, 53]. Parameters P , η_e , C_a , and C_b are identical to other three beams. The core radius and the two ring radii are denoted by r_{G_0} , r_{R_1} , and r_{R_2} , with corresponding ring thickness r_{T_1} , and r_{T_2} . Numerical values of these parameters are listed in Table 2. The visual representation of the Bessel beam is provided in the fourth column of Figure 3, showing its 3D, top and transverse profiles in (a), (b), and (c), respectively.

$$I_B = \frac{C_a a_0 P \alpha}{\pi r_{G_0}^2} \exp \left[-C_b \left(\frac{r^2}{r_{G_0}^2} \right) \right] + \frac{C_a a_1 P \alpha}{\pi r_{T_1}^2 Y(r_{R_1}, r_{T_1})} \exp \left[-C_b \left(\frac{(r - r_{R_1})^2}{r_{T_1}^2} \right) \right] + \frac{C_a a_2 P \alpha}{\pi r_{T_2}^2 Y(r_{R_2}, r_{T_2})} \exp \left[-C_b \left(\frac{(r - r_{R_2})^2}{r_{T_2}^2} \right) \right] \quad (17)$$

Table 2: Laser processing parameters.

Coeff.	Gaussian	Flat-Top	Ring	Bessel	Unit
P	25	25	25	25	mW
C_a	2	2	2	2	
C_b	2	2	2	2	
η_e	0.75	0.75	0.75	0.75	
r_G	70	—	—	—	μm
r_{FT}	—	70	—	—	μm
k	—	4.2	—	—	
r_R	—	—	70	—	μm
r_T	—	—	20	—	μm
r_{G_0}	—	—	—	22.65	μm
r_{R_1}	—	—	—	40	μm
r_{T_1}	—	—	—	30	μm
r_{R_2}	—	—	—	70	μm
r_{T_2}	—	—	—	20	μm
a_0	—	—	—	0.1	
a_1	—	—	—	0.5	
a_2	—	—	—	0.4	

6 Material Properties

Table 3: Temperature dependent material properties of individual phases.

Properties	Unit	Phase	Expression	Reference
Density (ρ)	kg/m^3	LIQUID	$19325.28 - 1.44T$	[54]
		FCC	$19657.60 - 1.20T$	[54]
Specific Heat (Cp)	J/kg K	LIQUID	$158 + 5.08 \times 10^{-7} \exp(-0.0114T)$	[55]
		FCC	$132 + 2.5 \times 10^{-5} T^2 - 0.011T$	[55]
Thermal Conductivity (k_{th})	W/m K	LIQUID	$100 + 0.027397T$	[54]
		FCC	$338.91 - 0.0693T$	[54]
Phase Mobility (M)	m^5/Js	LIQUID	$2.90 \times 10^{-15} \exp \left(\frac{-3043}{8.31T} \right)$	[10]
		FCC	$4.88 \times 10^{-21} \exp \left(\frac{-3221}{8.31T} \right)$	[10]

In laser-induced phase-field simulations, the material properties of each phase play a critical role in determining phase evolution and microstructural development. Parameters such as phase mobility, thermal conductivity, density, and specific heat capacity directly influence the kinetics of phase transformation. Since these properties vary with temperature, assigning constant values in a non-isothermal laser processing model can lead to significant inaccuracies. In this work, temperature-dependent thermodynamic expressions are used to define the properties of a unary Au system.

The material parameters are specified for the two phases, LIQUID and FCC, and interpolated between them using the function described in Equation 1. Table 3 lists the temperature-dependent expressions for density (ρ), specific heat (C_p), thermal conductivity (k_{th}), and phase mobility (M). Thermal conductivities of both phases are assumed isotropic. The free energy, which acts as the driving force for phase transformation, depends solely on temperature for this unary system. Two formulations are considered: a fitted (fine-tuned) expression and one obtained from a thermodynamic database (TDB), as shown in Equation 13. To represent the diffuse interface between solid and liquid regions, a mushy zone is introduced. The temperature-dependent viscosity of the LIQUID phase (μ_{LIQ}) and the adjusted viscosity of this interface region μ_{adj} are both defined in Equation 18 [56].

$$\begin{aligned}\mu_{LIQ} &= \exp\left(-0.1990 + \frac{2669}{T}\right) \\ \mu_{adj} &= 10 \times \mu_{LIQ} \times [h_{LIQ} \times (1 - h_{LIQ})]\end{aligned}\tag{18}$$

The adjustment factor $[h_{LIQ} \times (1 - h_{LIQ})]$ peaks at $\eta = 0.5$, representing a 50% solid-liquid mushy zone, which enhances numerical stability and captures viscous damping in the diffuse interface. This formulation is consistent with phase-field models for alloy solidification, where such interpolations reproduce experimental morphologies and mushy zone thicknesses [18]. Additionally, temperature-dependent surface tension at the meltpool surface gives rise to the Thermo-Capillary Convection (Marangoni effect). The governing relation is presented in Equation 19, following Cairncross et al. [57], where H is the mean surface curvature, σ_{LIQ} the Cauchy stress tensor for the LIQUID phase at the free surface, γ_{ST} the surface tension, and $\nabla_s = (I - \vec{n}\vec{n}) \cdot \nabla$ is the surface gradient operator representing the Marangoni effect. This expression is implemented as a boundary condition in the MOOSE Framework. The second part of Equation 19 defines the temperature-dependent surface tension of gold given by Aqra et al. [58] also visualized in Figure 2, where ϕ denotes the area occupied by a single atom ($7.74 \times 10^{-16} \text{cm}^2/\text{atom}$), k is the Boltzmann constant, V_{melt} is the molar volume of gold at its melting temperature, and $V(T) = M_{mol}/\rho(T)$ represents the temperature-dependent molar volume, with M_{mol} being the molar weight. E_s denotes the sublimation energy (368 kJ/mol).

$$\begin{aligned}\vec{n} \cdot \sigma_{LIQ} &= -2H\gamma_{ST}\vec{n} - \nabla_s \gamma_{ST} \\ \gamma_{ST} &= \phi^{-1} \left(\frac{V_{melt}}{V(T)} \right)^2 kT \left[0.139 \frac{E_s}{RT} - 0.053 \right]\end{aligned}\tag{19}$$

7 Machine Learning and Digital Twin

The integration of experiments with mathematical models in real time has enabled advanced process control in various industries [59]. For additive manufacturing, this creates a need for computational models that can operate alongside experiments, providing process-property insights at a comparable speed. Phase-field FEM simulation remains one of the most robust methods for predicting mesoscale microstructure evolution and phase transitions. When coupled with multiphysics modules such as heat transfer and fluid flow, the model can capture the dynamic behavior of material characteristics, including the influence of temperature and phase on grain growth, meltpool flow velocity, and thermal-phase diffusion. However, despite the high physical fidelity of phase-field simulations, their computational cost and time requirements make real-time prediction and digital twin implementation impractical. Relying solely on FEM calculations for real-time phase prediction or closed-loop feedback in additive manufacturing is currently unrealistic due to the immense computational demand. A more practical approach lies at the intersection of predictive accuracy and computational efficiency, using a machine learning model trained on simulation-generated data to drive Digital Twin applications.

This work outlines a framework for constructing a Digital Twin framework. It demonstrates a conceptual model capable of monitoring experimental setups and tracking variables in real time, including meltpool geometry, its area, phase transformations, flow velocity and the interaction between thermal and phase diffusion. With surface temperature data, such as from an infrared thermal camera as input, the trained ML model can predict meltpool area, solid-liquid phase distribution, and internal flow velocity. To illustrate the concept, we developed a virtual Digital Twin prototype capable of running on a *Raspberry Pi*. The system processes a temperature distribution (in *numpy* tensor format) as input to the ML model. For this virtual setup, the temperature data is generated from a separate laser simulation without phase-field and fluid-flow physics. The trained ML model then predicts the corresponding phase microstructure, meltpool region, and flow velocity field. The outputs, input temperature, predicted phases, area data, and velocity fields are displayed through an interactive web application. A quantitative comparison of the computational cost of FEM-based phase-field

simulations versus ML inference, together with the ML training details, is provided in the [Appendix D1](#) to clarify the efficiency gains relevant for real-time deployment. The web app is designed for easy updates and future expansion to multiple alloy systems by simply retraining the ML model as more simulation data becomes available. This modular approach allows rapid adaptation of the Digital Twin to different materials and processing condition. Future work will extend the model’s training dataset to include laser processing of multi-component alloys at various scales and laser parameters (power and scan-speed), advancing towards a comprehensive multi-material Digital Twin for Additive Manufacturing.

7.1 Training Data and Model Architecture

In this study, data were generated from multiple FEM simulations by varying both the laser beam profile and the free energy expression. From eight distinct phase-field laser simulations, described in Sections 4 and 5, a total of 3220 observations of the temperature distribution and corresponding phase evolution, as well as the meltpool phase’s velocity, were extracted from the exodus file produced by the FEM solver. These data were converted into a three-dimensional data tensor with dimensions (*x-coordinate, y-coordinate, time*), containing numerical values of temperature, phase, and velocity at each spatial and temporal point. Two U-Net architecture based machine learning models were developed: ‘**PhaseNet**’ for predicting the phase and its corresponding meltpool area, and ‘**VelNet**’ for predicting flow velocity within the meltpool. The hyperparameters used in each of these models are tabulated in Table 4. Both PhaseNet and VelNet were trained using datasets generated from simulations under all four laser heat source types (Gaussian, Flat-Top, Ring, and Bessel). This design ensures that the models are generalizable and applicable to any of the laser heating conditions studied.

7.1.1 PhaseNet Model: Meltpool Morphology

The first model was developed to predict the phase distribution based on the temperature, and **so it can be considered as the architecture required to set up digital twin (MicroTwin) for quantifying 3D printed microstructure in real time**. The architecture of MicroTwin is represented by the portion within the red colored box (PhaseNet) in Figure 4. This stage aims to identify the solid and meltpool regions at each spatial point as the temperature evolves during the process. Given that the simulation data are represented in a 2D grid structure similar to images and phase classification is binary (0 or 1), the U-Net’s superior performance in image segmentation tasks makes it an appropriate choice for this application. The model takes a single-channel temperature grid as an input, and the output is a single-channel phase grid. The architecture of the model is represented in Figure 4 within PhaseNet block. The architecture comprises of series of encoders and decoders with a bottleneck in between. The encoder pathway has three downsampling stages, each consisting of a 3×3 convolutional layer with batch normalization and *ReLU* activation function, followed by 2×2 pooling. The feature channel along the encoder increase from 32 to 64 and then to 128 across the 3 stages, while the spatial resolution decreases by half in each stage. At the bottleneck, there are 256 channels, and the spatial dimension is reduced to 1/8th of that of the original dimension. The decoder pathway then reconstructs the original spatial and feature map. It consists of three upsampling stages using 2×2 transposed convolution. At each stage, we have skip connections that bridge the encoder and decoder, allowing the decoder to recover important spatial information that might have been lost during the downsampling. The final layer outputs a raw phase logit that is passed through a sigmoid activation function to decode the phase values between 0 and 1. These values then classified into either solid or liquid based on the threshold of 0.5 as shown in the equation below:

$$\text{phase}(i, j) = \begin{cases} 1, & \text{if } \sigma(z_{i,j}) > 0.5, \\ 0, & \text{otherwise,} \end{cases} \quad (20)$$

where $\sigma(\cdot)$ denotes the sigmoid activation and $z_{i,j}$ is the raw output at pixel (i, j) . The threshold ensures a separation between the solid and liquid phases, producing binary values of 0 and 1, which better reflects the actual physical state than the continuous probabilities.

7.1.2 VelNet Model: Automated Prediction of Kinematic Feature

The second model predicts the velocity field **using the temperature distribution externally and the phase information predicted by the PhaseNet Model internally via the MicroTwin**. This VelNet Model utilizes the same U-Net architecture, but with two input channels: one corresponding to temperature and the other to phase, and a single output channel representing the meltpool velocity. The VelNet architecture is the backbone software library for constructing the **FlowTwin** in additive manufacturing process. The VelNet architecture’s dependence on PhaseNet for Physics-informing

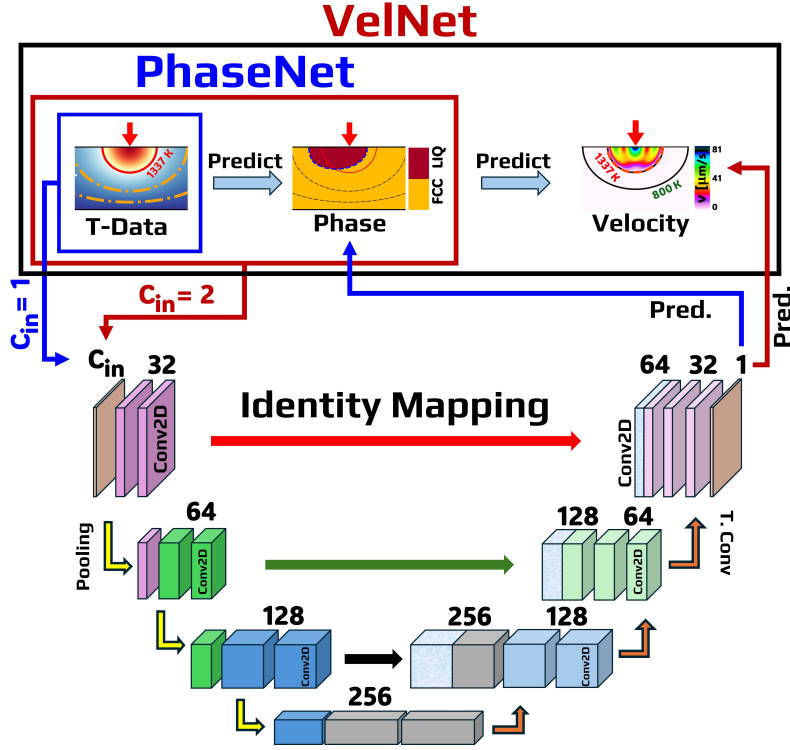


Figure 4: Schematic of the customized multi-step U-Net architectures developed for predicting phase evolution and meltpool velocity. The PhaseNet model predicts the phase distribution from a single-channel temperature input (T-Data, $C_{in} = 1$), while the VelNet model predicts the meltpool velocity using two input channels, temperature (T-Data) and the PhaseNet-predicted phase ($C_{in} = 2$). Both models share the same encoder-decoder structure, consisting of convolutional and pooling layers that extract hierarchical spatial features. The encoder compresses spatial dimensions while increasing feature channels (C_{in} , 32, 64, 128), and the decoder restores spatial resolution through upsampling and skip connections (lateral arrows), reducing channels from 256 to 1. The primary difference between the two models lies in their training objectives: PhaseNet uses a binary cross-entropy loss for phase classification (LIQUID vs. FCC), whereas VelNet employs a mean squared error (MSE) loss for velocity regression, combined with a hard-pass filter to constrain predictions within the LIQUID phase region. The PhaseNet and VelNet programs are loaded on the Raspberry Pi microcomputer (refer Figure 13) to render it as a physics-informed conditionally automated device.

the domain with phase mask implies that FlowTwin is built on top of MicroTwin. In other words, the MicroTwin is hierarchically/vertically linked to FlowTwin in terms of information flow.

The velocity prediction network shares the same U-Net encode-decoder framework as the PhaseNet model, as shown in Figure 4, with modifications to accommodate two input channels performing regression rather than classification. The FlowTwin is represented by the black colored box (VelNet) in Figure 4 and it is clear therein that it totally sits on top of MicroTwin. Specifically, the sigmoid activation function and binary cross-entropy loss were replaced with a linear output layer and mean squared error loss, respectively. However, there is another key modification in the output stage. We introduce a hard mask based on the laws of physics on the output velocity. As the velocity in the solid phase regions can be treated as non-existent (since they are nearly equal to zero), we multiply the final predicted velocity by the binary phase mask as follows:

$$v_{\text{pred}}(i, j) = \text{phase}(i, j) \times \hat{v}(i, j) \quad (21)$$

where $\hat{v}(i, j)$ is the VelNet model's predicted output velocity. This masking operation ensures that velocities are driven to zero wherever the phase (order-parameter) equals zero, which is represented by the solid FCC phase in the domain.

The complete model implementation, including encoder, bottleneck, and decoder blocks with their respective convolutional, max pooling, and upsampling layers, is available on the [PhaseVelNet GitHub](#) page, as referenced in Section 13.

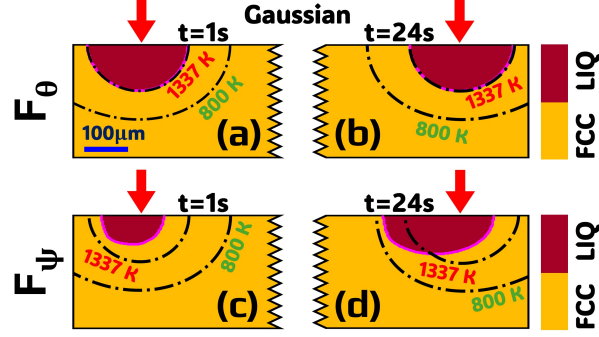


Figure 5: Phase evolution in a unary Au system irradiated by a Gaussian laser beam (25 mW , $30\text{ }\mu\text{m/s}$) scanning from the top left to the top right of the computational domain. Subplots (a) and (b) shows phase-field simulations using the fine-tuned free energy expression (F_θ) at $t = 1\text{ s}$ and $t = 24\text{ s}$, respectively, where the LIQUID phase (dark red) aligns with the melting temperature contour of 1337 K . Subplots (c) and (d) presents results using the TDB-based free energy expression (F_ψ), indicating a time lag between thermal and phase diffusion, with thermal diffusion leading. At $t = 1\text{ s}$, the melt pool is $18484.18\text{ }\mu\text{m}^2$ for F_θ and $7575.42\text{ }\mu\text{m}^2$ for F_ψ , which converge to an average of $\approx 17800\text{ }\mu\text{m}^2$ by the end of the simulation, though the melt pool shape differ significantly. The results for Flat-Top, Ring and Bessel beam are provided in Figure A2 of Appendix A

8 Result and Discussion

This section presents the results obtained from phase-field simulations and machine learning model predictions. In Section 8.1, we analyze the thermo-phase response corresponding to different free energy expressions used during laser-induced simulations. Section 8.2 discusses the evolution of the melt pool, its area, and temperature distribution across the computational domain under various laser beam profiles. Finally, Section 8.4 presents the outcomes of the machine learning model trained on simulation data, demonstrating its capability to predict laser-induced multiphase behavior. We also outline the use of temperature distribution data tensors for rapid prediction of phase transformations through the trained ML models, along with the conceptualization of a virtual digital twin framework.

8.1 Effects on Phase Response by Driving Force Formulations

Phase evolution is primarily governed by the system’s free energy, the phase with the lowest free energy at a given temperature determines temporal stability. This study compares the phase response, stability, and transition behavior during non-isothermal laser processing using two Gibbs free energy formulations that dictates the phase-field dynamics. As shown in Graph 2, for the fine-tuned expression 2(a), the free energy curves for the two phases are parabolic and intersect near the melting point of Au, $T_{\text{melt}} = 1337\text{ K}$, forming a small energy gap. In contrast, the TDB-based free energy curves (Figure 2(b)) intersects directly at T_{melt} without a visible gap between the LIQUID and FCC phases.

For the LIQUID phase evolution using the fine-tuned expression (F_θ), the results differ distinctly from those with the TDB-based expression (F_ψ), as seen in Figure 5. The initial LIQUID region ($2500\text{ }\mu\text{m}^2$) grows into a semi-circular melt pool that follows the 1337 K contour at around 1 s of simulation time (Figure 5(a)). The melt pool boundary continues to trace this contour until the heat source is turned off Figure 5(b). In contrast, for F_ψ (Figure 5(c)), the melt pool is not fully developed at 1 s , and the 1337 K thermal front leads the actual LIQUID front. This creates a

Table 4: Model Hyperparameters

Hyperparameter	PhaseNet	VelNet
Batch Size	16	16
Learning Rate	1×10^{-3}	1×10^{-3}
Number of Epochs	200	200
Input Channels	1	2
Output Channels	1	1
Loss Function	BCEWithLogitsLoss	MSE
Optimizer	Adam	Adam
Test Split	15%	15%
Validation Split	15%	15%

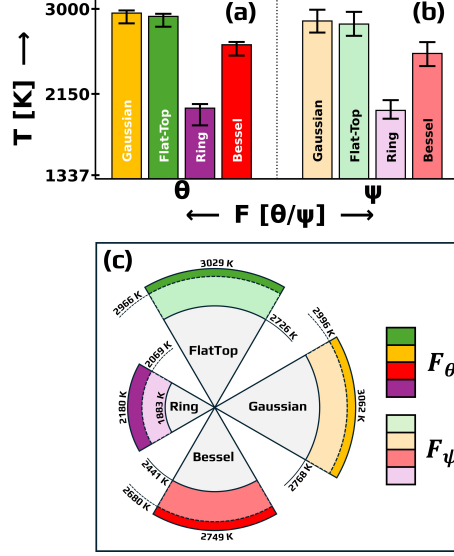


Figure 6: Bar graph comparing the average peak temperature at the top surface of the computational domain as the laser scans from left to right during phase-field simulations. In (a), results are shown for simulation using fine-tuned free energy expression (F_θ), where the Gaussian beam reaches the highest peak temperature of 3062 K, followed by Flat-Top (3029 K), Bessel (2749 K), and Ring (2180 K) profiles. In (b), the results using the TDB-based free energy expression (F_ψ) shows a similar trend, with peak temperatures of 2996 K, 2966 K, 2680 K, and 2069 K for Gaussian, Flat-Top, Bessel, and Ring beams, respectively. Overall, simulations using F_θ yield consistently higher temperature values across all beam profiles compared to those using F_ψ . (c) The bird-eye overview of maximum temperature for four laser beam types with both free energy formulations in a radar chart.

physical separation between the temperature contour and the meltpool boundary, as shown in Figure 5(c) and (d), with red contours indicating the meltpool and black dotted lines the temperature field. This positional lag between thermal and phase diffusion, referred to as lead-lag dynamics described in detail in our previous work [10], is observed for F_ψ but not for F_θ . While Figure 5 illustrates results for the Gaussian beam, similar distinctions in LIQUID phase evolution are observed for all other laser beams.

Another key difference between F_θ and F_ψ appears in the peak temperature range across the domain. Figure 6 shows the variation of maximum surface temperature for both energy models: F_θ (Figure 6(a)) consistently yield higher peak temperature than F_ψ (Figure 6(b)). For instance, under the Gaussian beam, F_θ produces an *average peak temperature* of 2958 K, compared to 2880 K for F_ψ . moreover, F_ψ exhibits a wider range of temperature fluctuations over time, as reflected by the larger error bars across all beam profiles. Similarly, Figure 6 (c) presents the bird-eye view of the *maximum temperatures* attained by the four laser beam profiles with variations in two free expressions in a radar-chart.

Free energy formulation also influences meltpool flow velocity. Figure 7(a) and (b) shows the velocity fields for F_θ under Bessel and Ring at $t = 12.5$ s while, Figure 7(c) and (d) shows the same for F_ψ . For F_θ , the peak velocities are $81.17 \mu\text{m/s}$ (Bessel) and $18.90 \mu\text{m/s}$ (Ring). For F_ψ , these values are $56.29 \mu\text{m/s}$ and $23.71 \mu\text{m/s}$ respectively. The Ring beam is an exception, where the TDB based model (F_ψ) yields a higher flow velocity than (F_θ), a behavior discussed further in Section 8.2.

Another observation in simulation with F_ψ is the asymmetry in temperature distribution: the leading region (right of the moving laser) shows a wider thermal spread than the trailing region (left side). In Figure 8(c) and (d), the spatial gap between the 1337 K and 800 K contours at the leading surface (β_ψ and β'_ψ) is larger than at the trailing surface (α_ψ and α'_ψ). In contrast, for F_θ , these contours remains nearly symmetric ($\alpha_\theta \approx \beta_\theta$ and $\alpha'_\theta \approx \beta'_\theta$) as shown in Figure 8(a) and (b).

8.2 How Heat Source Type Alters the Morphology

This section discusses how four different laser beam profiles influences phase dynamics in the phase-field simulations. Starting with the temperature distribution, it is observed that the Gaussian beam produces the highest peak temperature, followed by the Flat-Top beam. The domain reaches a peak of 3062 K and 2996 K for F_θ and F_ψ respectively under the

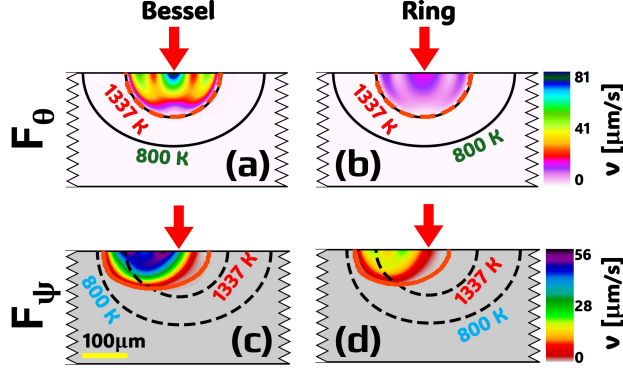


Figure 7: Subplots showing the melt pool velocity profiles when the laser beam reaches the domain midpoint ($500 \mu m$ from the top-left corner) at $t = 12s$ for two laser beam profiles under two free energy expressions. In (a) and (b), the velocity distribution within the LIQUID phase is shown for Bessel and Ring beams using the fine-tuned free energy expression (F_θ). Similarly, (c) and (d) present results for the same beam profiles using the TDB-based expression (F_ψ). The melt pool velocity is consistently higher for the Bessel beam compared to the Ring beam across both free energy cases. For F_θ , the maximum velocities are approximately $81 \mu m/s$ (Bessel) and $18.90 \mu m/s$ (Ring), while for F_ψ , the respective maxima are $56 \mu m/s$ and $23.71 \mu m/s$. Comparing the two energy expressions, F_θ yields higher flow velocities for the Bessel beam, whereas F_ψ produces greater melt pool flow for the Ring beam. The plots for Flat-Top and Gaussian beam are provided in Figure A3 of Appendix A

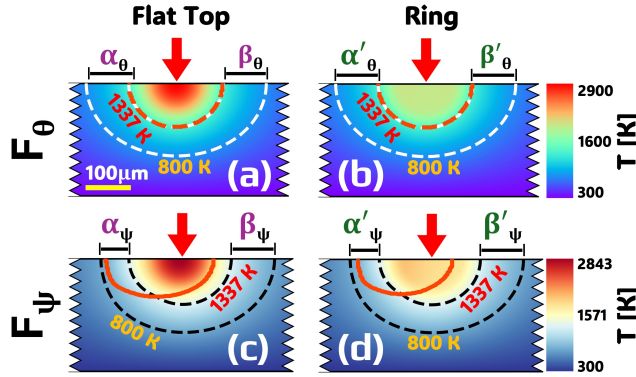


Figure 8: Comparative illustration of temperature distribution for Flat-Top and Ring laser beam profiles at $t = 12.5s$, when the laser reaches the midpoint of the top surface of the computational domain. Subplots (a) and (c) correspond to the Flat-Top beam under fine-tuned (F_θ) and TDB-based (F_ψ) free energy expressions, respectively, while (b) and (d) represent the Ring beam under the same conditions. For F_θ , both Flat-Top and Ring profiles show nearly symmetrical temperature contours at leading and trailing edges ($\alpha_\theta \approx \beta_\theta \approx 97.76 \mu m$) and ($\alpha'_\theta \approx \beta'_\theta \approx 92.27 \mu m$), with peak temperature of $2900 K$ and $1967 K$, respectively. The $1337 K$ contour aligns precisely with the melt pool boundary (dashed red line), indicating no lag between phase and thermal fronts. In contrast, simulations with F_ψ show a clear offset between temperature and phase fronts (red and black dashed lines) and asymmetry in the contour spacing. For Flat-Top, the leading edge gap ($\beta_\psi = 94.76 \mu m$) exceeds the trailing one ($\alpha_\psi = 62.34 \mu m$), and a similar trend appears for the Ring beam ($\beta'_\psi = 94.76 \mu m$, $\alpha'_\psi = 64.83 \mu m$). Overall, the Flat-Top beam produces higher temperature across the domain compared to the Ring beam, irrespective of the free energy expression.

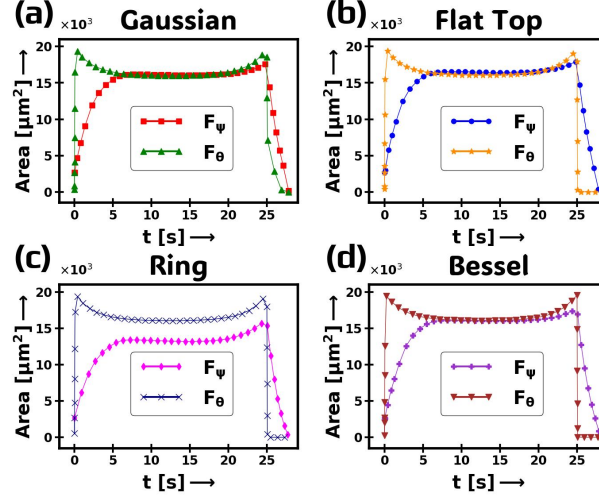


Figure 9: Plots illustrating the evolution of meltpool (LIQUID) phase area over the simulation duration for four laser beam profiles and two free energy expressions. In (a), the Gaussian beam shows that the fine-tuned free energy (F_θ) causes a rapid rise in meltpool area to $19422.59 \mu m^2$, stabilizing around $16497.74 \mu m^2$, while the TDB-based expression (F_ψ) produces a gradual growth reaching about $15551.55 \mu m^2$ until the laser is turned off at $t = 25s$. Similar behavior is seen for the Flat-Top (b) and Bessel (d) beams, both averaging near $16500 \mu m^2$, with F_θ showing an initial sharp rise and F_ψ a smoother increase. The Ring beam (c) deviates from this pattern: the F_θ simulation maintains an average meltpool area of $16549.60 \mu m^2$, while the F_ψ simulation shows a smaller area averaging $12967.09 \mu m^2$ and peaking near $16338.35 \mu m^2$. A distinct delay is observed in meltpool formation and solidification for F_ψ , contrasting with the sharper transitions seen in F_θ .

Gaussian beam, while the Flat-Top beam attains $3029 K$ and $2966 K$ (Figure 6(a) and (b)). For the Bessel beam, the maximum temperature are $2749 K$ (F_θ) and $2680 K$ (F_ψ). Although these three beam produces comparable temperature range (above $2500 K$), the Ring beam exhibits a distinct thermal response. Its peak temperature are notably lower, $2180 K$ and $2069 K$ for F_θ and F_ψ respectively, due to the unique way it distributes heat. Unlike Gaussian, Flat-Top and Bessel beams that deliver concentrated irradiance at the center, the Ring beam has near-zero central intensity, focusing energy along a circular ring. The resulting bidirectional heat flow- both inward and outward from the ring, creates a more uniform temperature field and prevents sharp temperature spikes. This behavior is clearly seen in Figure 8(b) and (d), where the Ring beam produces a smoother, radially symmetric temperature profile compared to the more peaked distribution of the Flat-Top beam in Figure 8(a) and (c). The temperature fields for Gaussian and Bessel beams at mid-simulation are shown in Figure A4 of Appendix A.

A similar, influence of beam shape is reflected in the evolution of the LIQUID phase. The meltpool area, tracked at each time step, is summarized in Figure 9 for all beam types and both free energy expressions. For all beams except the Ring, the meltpool area evolves similarly in both F_θ and F_ψ simulations and converges by mid-simulation. However, for the Ring beam, the meltpool area with F_ψ remains significantly smaller than with F_θ . From these results, one can infer that beam shape directly impacts meltpool size and heat-affected zone (HAZ), guiding laser selection for specific additive manufacturing processes. For example, the Flat-Top beam produces the largest average meltpool area $16545.82 \mu m^2$ (F_θ) and $15839.72 \mu m^2$ (F_ψ), making it suitable for processes requiring larger meltpools or broader HAZ. Conversely, the Ring beam, with its uniform but smaller heating region, is more suitable where consistent temperature distribution and tighter HAZ control are desired.

The differences in temperature evolution across beam profiles are analyzed along the horizontal midline of the domain, indicated by the red arrow in Figure 10. In all cases, temperature increases as the laser approaches the midpoint, peaks when the beam center passes over it, and decreases thereafter. Figure 10(a) compares Gaussian and Ring beams; Figure 10(b) compares Gaussian and Bessel; Figure 10(c) compares Ring and Flat-Top; and Figure 10(d) compares Flat-Top and Bessel. The Gaussian beam consistently produces higher peak temperature than other beams. In Figure 10(c), the Flat-Top beam heats the midpoint to $2843.02 K$, while the Ring beam maintains a nearly constant temperature of $1958.09 K$ between $t = 10.61 s$ to $14.39 s$. This flattened thermal response of the Ring beam reflects steady and uniform heating, with gradual temperature rise and fall due to the ring's leading and trailing edges. Overall, Gaussian, Flat-Top, and Bessel beams produce a bell-shaped temperature profile at any surface point, while the Ring beam yields a steady, plateau-like profile.

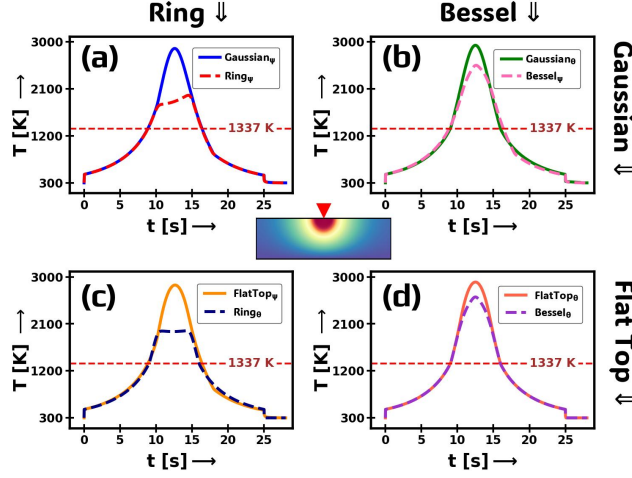


Figure 10: Time-temperature plots for a spatial point located at the mid-top surface (inset). Subplot (a) compares Gaussian and Ring beams with F_ψ , showing a bell-shaped temperature curve for the Gaussian beam peaking at 2873.29 K, while the Ring beam exhibits a flatter profile with a peak of 1975.98 K. In (b), the Gaussian beam with F_θ (green) reaches 2933.98 K, higher than the Bessel beam with F_ψ (pink, 2550.24 K). Subplot (c) compares Flat-Top (F_ψ) and Ring (F_θ) beams, where the Ring beam provides uniform heating (peak 1958.09 K) and the Flat-Top beam shows gradual heating and cooling (peak 2873.02 K). In (d), both Flat-Top and Bessel beams with F_θ yield bell-shaped curves peaking at 2900.08 K and 2614.45 K, respectively. Readers are referred to Figure 6c to obtain a bird-eye overview of the overall impact with different free energy expressions for each heat source types.

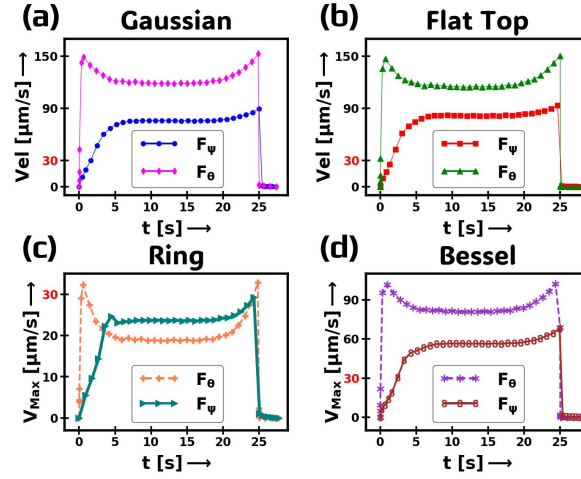


Figure 11: Subplots showing the evolution of maximum melt pool velocity during simulation for different laser beam profiles and free energy expressions. In (a), the Gaussian beam produces a higher peak velocity with F_θ (magenta line with diamond markers) than with F_ψ (blue line with circular markers), averaging 123.75 $\mu\text{m/s}$ and 71.76 $\mu\text{m/s}$, respectively. A similar trend is seen for the Flat-Top beam in (b), with average peak velocities of 119.94 $\mu\text{m/s}$ (F_θ) and 76.86 $\mu\text{m/s}$ (F_ψ). In contrast, for the Ring beam ((c)), F_ψ yields slightly higher flow (22.65 $\mu\text{m/s}$) than F_θ (20.37 $\mu\text{m/s}$). For the Bessel beam ((d)), the trend reverts to higher values with F_θ (84.65 $\mu\text{m/s}$) compared to F_ψ (52.39 $\mu\text{m/s}$). Note that the y-axis scale differs in (c) and (d) to account for the lower velocity range; a red marker at 30 $\mu\text{m/s}$ is included across all subplots for reference.

For simulations using the fine-tuned free energy (F_θ), the beam-induced meltpool flow velocity follows the temperature trends. The Gaussian beam generates the highest average flow velocity of $123.75 \mu\text{m/s}$ (magenta line, Figure 11(a)), followed by the Flat-Top beam at $119.94 \mu\text{m/s}$ (green line, Figure 11(b)). The Bessel beam reaches a maximum of $84.65 \mu\text{m/s}$ (purple line, Figure 11(d)), while the Ring beam produces the lowest velocity of $20.37 \mu\text{m/s}$ (cyan line, Figure 11(c)). For the TDB-based energy model (F_ψ), the order of the flow velocity shifts slightly, Flat-Top records the highest ($76.86 \mu\text{m/s}$), followed by Gaussian ($71.76 \mu\text{m/s}$), Bessel ($52.39 \mu\text{m/s}$) and Ring ($22.65 \mu\text{m/s}$), shown by red, blue, brown, and orange curves in Figure 11(b), (a), (d) and (c) respectively. As also seen in the temperature data, the Ring beam remains an exception, its peak velocity is slightly higher in F_ψ ($22.65 \mu\text{m/s}$) than in F_θ ($20.37 \mu\text{m/s}$), consistent with its uniform and stable heating characteristics.

8.3 Uncertainty reduction in T-magnitudes

In laser AM, uncertainty in the prediction of peak meltpool temperatures is one of the major factors that limits build quality and process reliability. When thermodynamic models overestimate the temperature, often by several hundred kelvins, the melt pool becomes excessively hot. For metals such as Au or Ti-6Al-4V, temperatures above roughly 2000 K to 3000 K can lead to keyholing, where intense vapor recoil forms deep pores, spallation defects, and localized evaporation of low-boiling alloying elements. These effects alter the local chemistry, promote brittle phases, and ultimately compromise mechanical performance. If the model underpredicts the temperature, the opposite problem arises: insufficient melting causes lack-of-fusion defects, delamination between layers, and strongly anisotropic microstructures. These defects serve as preferred sites for fatigue cracking and can reduce component life by more than an order of magnitude. Even moderate temperature errors (100-200 K) can change melt-pool shape by 20-30%, leading to irregular layer bonding and high build-failure rates in production environments. Additional consequences include wasted energy due to overheating and safety concerns from high-velocity spatter. Our TDB-based free-energy function, F_ψ , obtained directly from CALPHAD-consistent thermodynamic databases, substantially reduces these uncertainties. Compared with the empirically fitted free-energy expression F_θ , the TDB-based model predicts 60-100 K lower and more realistic peak temperatures as shown in Figure 6.

The ramp-up rate of driving force for phase transition (Λ) near the melting temperature ($\pm\delta T_m = 5 \text{ K}$) is small for TDB based formulation ($\Lambda_\psi = -9.511 \text{ J/mol/K}$) whereas, it is inflated by 81 times for fitted expression with $\Lambda_\theta = -770.45 \text{ J/mol/K}$ as depicted in Figure A1 of Appendix A. This implies that during laser heating the fitted free energy based phase field model directly transforms FCC into LIQUID phase with a significantly sufficient barrier to resolidify, when the melting regime is reached and the temperature increases in the LIQUID phase faster. On the other hand, it might be possible for the material within the melting regime to remain in equilibrium between FCC and LIQUID phases in context of the model using TDB expressions.

The temperature-slope of the thermodynamic driving force for the solid-liquid transition, ($\Lambda = d\Delta F/dT$), differs sharply between the two free-energy formulations. In the TDB-based model this slope is modest ($\Lambda_\psi = -9.511 \text{ J/mol/K}$), while in the fitted expression it becomes **over 81 times steeper** ($\Lambda_\theta = -770.45 \text{ J/mol/K}$). Thermodynamically, this slope is tied to the entropy discontinuity at the melting point, ($\Delta S = L_f/T_m$), as the latent heat of fusion satisfies $L_f = \frac{d\Delta F}{dT}|_{T_m}$, with ($\Delta F = F_{\text{LIQ}} - F_{\text{FCC}}$) being the Gibbs free-energy difference near (T_m). A steeper ($|d\Delta F/dT|$) corresponds to a higher effective latent heat and forces the transformation across an extremely narrow temperature range. In the fitted model, this exaggerated slope causes the driving force ($|\Delta F|$) to shoot upward as soon as the material enters the melting regime. As a result, the phase field quickly converts FCC to LIQUID, and because (ΔF) continues to increase unnaturally fast in the LIQUID phase, the model strongly suppresses solid-liquid coexistence. This creates a thermodynamically overstabilized LIQUID, which heats up more rapidly, trapping thermal energy and promoting unrealistic thermal overshoots. In contrast, the TDB-based formulation yields a much smoother and physically grounded variation of ($\Delta F(T)$). Its latent-heat signature is correctly distributed across a finite temperature window, enabling genuine thermodynamic coexistence of FCC and LIQUID phases. Sustained presence of high-conductivity solid near the melting point. Because the transition is not artificially compressed, a fraction of solid persists longer during heating. Solid gold has a thermal conductivity of $246.3945 \text{ W/m} \cdot \text{K}$ at $T = 1335 \text{ K}$, more than that of the LIQUID ($136.71 \text{ W/m} \cdot \text{K}$ at $T = 1340 \text{ K}$). The continued presence of the solid therefore enhances radial heat extraction, preventing localized heat accumulation and a broader latent-heat absorption interval. In the TDB formulation, the latent heat is absorbed over a physically meaningful temperature range rather than at an artificially sharp point. This effectively buffers incoming laser energy, delaying the temperature rise and flattening the thermal trajectory. The material spends more time in the mixed-phase regime where enthalpy is absorbed without a rapid temperature increase. Together, these thermodynamic effects result in 60-100 K lower peak temperatures in the TDB-based simulations compared with the fine-tuned model. The fine-tuned expression, by overstating the driving force slope, collapses the phase transition into a narrow band, suppresses the high-conductivity solid prematurely, and eliminates latent-heat buffering conditions that all amplify thermal overshoot. Conversely, the TDB formulation enforces CALPHAD-consistent phase equilibria, distributes latent heat realistically, preserves conductive solid for

longer, and prevents artificial trapping of heat in low conductivity mushy zones. This combination stabilizes the thermal response, yielding peak temperatures far closer to experimentally observed values in laser melting of Au.

Ultimately, the strength of the TDB-based formulation is in providing thermodynamically consistent and reliably constrained temperature predictions. By embedding the correct latent-heat behavior, preserving realistic solid-liquid coexistence, and avoiding the artificial steepening of the driving force seen in the fitted model, the TDB approach delivers temperature fields that adhere closely to CALPHAD-based phase equilibria and thus markedly reduce the error margin in peak-temperature estimation. This reduction in uncertainty is critical for laser AM: accurate thermal predictions ensure that the process window is identified correctly, preventing inadvertent transitions into overheating regimes associated with keyholing, vapor-driven porosity, solute loss, and microstructural degradation, while also avoiding the under-melting conditions that cause lack-of-fusion defects and weak interlayer bonding. In contrast, the heuristically fitted free-energy expression injects substantial uncertainty into the temperature field-its exaggerated driving-force slope, compressed melting interval, and unrealistic thermal response amplify both over- and under-prediction risks. Such variability propagates directly into melt-pool geometry, defect formation, and build reliability. Therefore, the TDB-based formulation provides a more accurate, lower-uncertainty, and physically grounded foundation for predicting melt-pool temperatures, ultimately enabling more consistent, defect-resistant, and scalable additive manufacturing.

8.4 Learning the Meltpool Profile

The U-Net architectures predicting the microstructure phase and the meltpool flow velocity described in Section 7 were trained for 200 epochs on the data generated from FEM simulations. The training performance of both models are listed in Table 5. Both models exhibit good prediction capabilities across training, validation, and testing. PhaseNet achieved an accuracy score of 0.9992, 0.998, and 0.997 in training, validation and test dataset respectively. VelNet also showed similar performance with higher R^2 , and lower loss across training ($R^2=0.9994$, $MSE=0.045$), validation ($R^2=0.9992$, $MSE=0.060$) and testing ($R^2=0.9990$, $MSE=0.068$) indicating generalization capability and minimal overfitting.

Figure 12(a) shows the microstructural similarity index (μ_{SIM}) calculated from the LIQUID phase predictions across 483 test samples. The μ_{SIM} values consistently exceed 0.995, with most values approaching 0.999. This high level of similarity indicates that the model accurately captures phase distribution patterns from the given temperature field inputs. The calculation method for μ_{SIM} is described in Appendix B. The training behavior of PhaseNet is presented in Figure 12(b), which displays the binary cross-entropy (BCE) loss and classification accuracy over 200 training epochs. The model converged rapidly, with both training and validation accuracy reaching approximately 0.999 within the first 50 epochs. These accuracy values remained stable throughout the remaining training period, showing only minor variations. The loss values followed a similar trend, decreasing from an initial value of about 0.25 to near zero within 50 epochs, after which they remained consistently low for the duration of training. The performance of VelNet is shown in Figures 12(c) and (d). Figure 12(c) presents a parity plot that compares the predicted liquid phase velocities against the ground truth values for the test dataset. The predicted values align closely with the actual values along the diagonal, demonstrating the model’s generalization capability. Figure 12(d) shows the training performance of VelNet in terms of R^2 coefficient and mean squared error (MSE) loss. The training behavior of VelNet is similar to that observed for PhaseNet. Both the training and validation R^2 values rapidly increase to approximately 0.99 in the early epochs and stabilize near 1.0 for the remainder of the training period. The MSE loss curves show a corresponding pattern, with a sharp decrease in the initial epochs followed by convergence to near-zero values. Minor fluctuations in the loss values are observed primarily during the early training phase, after which the curves remain stable.

Table 5: Performance summary of PhaseNet (Phase Prediction) and VelNet (Velocity Prediction) models.

Dataset	PhaseNet			VelNet	
	Accuracy	BCE Loss	MSE	MAE	R^2
Training	0.9992	0.0016	0.045	0.062	0.9994
Validation	0.9989	0.0024	0.060	0.071	0.9992
Testing	0.9979	0.0060	0.068	0.079	0.9990

To further evaluate the model’s predictive capability, a separate FEM simulation was conducted in the MOOSE Framework using only the heat transfer physics module. This setup computed the temperature distribution under a Bessel laser beam profile with the same power and scan speed as in the base phase-field simulations. An interpolation function (Equation 22) was applied to smoothly vary material properties- thermal conductivity, specific heat, and density, across the domain relative to the melting temperature of gold (1337 K).

$$h_T = \frac{1}{1 + e^{-10\left(\frac{T}{T_{\text{melt}}} - 1\right)}} \quad (22)$$

The resulting temperature data were converted into tensor format and fed as input to the MicroTwin (refer Figure 13). The model-predicted melt pool phase evolution from the temperature history was compared against results from the full phase-field simulation. Figure 14(a) shows the temperature field from the heat transfer simulation, while Figure 14(b) presents the corresponding phase evolution from the full phase-field model. The predicted phase distribution from the trained PhaseNet model is shown in Figure 14(c), and the prediction error map in Figure 14(d). In this figure, green regions indicate *False Positives* (FP), where the model predicted the presence of melt pool phase absent in the phase-field result; red regions denote *False Negatives* (FN), where the model failed to identify the LIQUID phase; and light-yellow areas mark the *Correctly Predicted* (CP) regions where both results match. In addition, the LIQUID phase area at the presented time instant from the full phase-field FEM simulation was $17,363 \mu\text{m}^2$, whereas PhaseNet predicted $18,046 \mu\text{m}^2$, resulting in a melt pool area prediction error of about 3.93%. The temperature feature should be supplied as input feature whereas phase related features can be internally fed to the FlowTwin of Figure 13. When the phase related features are supplied to the FlowTwin via the MicroTwin, the features are the machine learned ones. Also, it is possible to supply both temperature and phase features externally as input to the micro-computer. In this case, the phase features must be obtained from the FEM model. The phase features supplied to VelNet model (FlowTwin) to result in Figure 14(e) were obtained from MicroTwin. Figure 14(e) shows the velocity field predicted by the VelNet model, while the corresponding full phase-field simulation result at the same time instance is presented in Figure 14(f). The absolute error between the predicted and simulated velocity fields is shown in Figure 14(g). For VelNet, the input variables were the temperature distribution from the stand-alone heat transfer simulation and the predicted phase data from PhaseNet. Thus, both the phase distribution and melt pool flow velocity can be inferred solely from the thermal distribution across the domain.

9 Virtual Digital Twin and Outlook

The integration of IoT and remote sensing with digital twins offers an effective approach to minimizing laser-related hazards in industrial and research applications. Through real-time monitoring and predictive capability, digital twins can simulate laser-material interactions under varying conditions, reducing the risk of thermal damage, structural defects, and safety issues. IoT-enabled sensors provide continuous data streams, allowing adaptive control of laser parameters and improving overall process reliability [60, 61].

Buliding on this, we propose a Digital Twin framework that employs training machine learning models to quantify and predict phase evolution, melt pool area, and velocity profiles in real time during additive manufacturing. The trained models can be deployed on a microcomputer such as a *Raspberry Pi*, connected to an infrared camera that records surface temperature data of the specimen. The model processes this temperature data and displays phase transformation details in real time.

To conceptualize this idea, a Virtual Digital Twin (VDT) [62] was developed with a web interface (WebApp), the working flow of this VDT is presented in Figure C1 of Appendix C. Instead of using infrared camera data, the virtual twin operates on temperature fields generated from a standalone laser heat-transfer simulation that excludes phase-field and fluid-flow physics (a Bessel beam is used as an example). The resulting temperature data are first passed to the trained PhaseNet model, which predicts the LIQUID and FCC phases and computes their repsective areas. These outputs, together with temperature data, are then supplied to the VelNet model, which infers the velocity field within the melt pool region. The WebApp also incorporates the original phase-field simulation results (used to train ML models) to compute and visualize real-time error between simulated and predicted values for both phase and velocity fields. The Virtual Digital Twin WebApp is accessible at: [VDT WebApp](#)

This virtual twin framework can be readily extended to a physical digital twin by deploying the trained ML models on a *Raspberry Pi*. In this setup, actual thermal data are captured via infrared camperas, processed into numerical arrays, and used directly as model input. The prediction scope can be expanded in future work by training on data from multi-material systems (multi-component alloys) across different laser powers, scan speeds, and length scales, thereby enabling a generalized multi-material Digital Twin model.

In this context, the phase-field-informed U-Net architectures: PhaseNet and VelNet, running on a *Raspberry Pi* with infrared input exemplify conditional automation in additive manufacturing. PhaseNet takes temperature data and predicts structural features representing melt pool morphology, while VelNet uses temperature and phase data to infer melt pool velocity. The process is conditional because model outputs depend directly on the physical state encoded in the phase-field-based inputs. These networks learn mappings between phase-field states and melt pool behavior, which

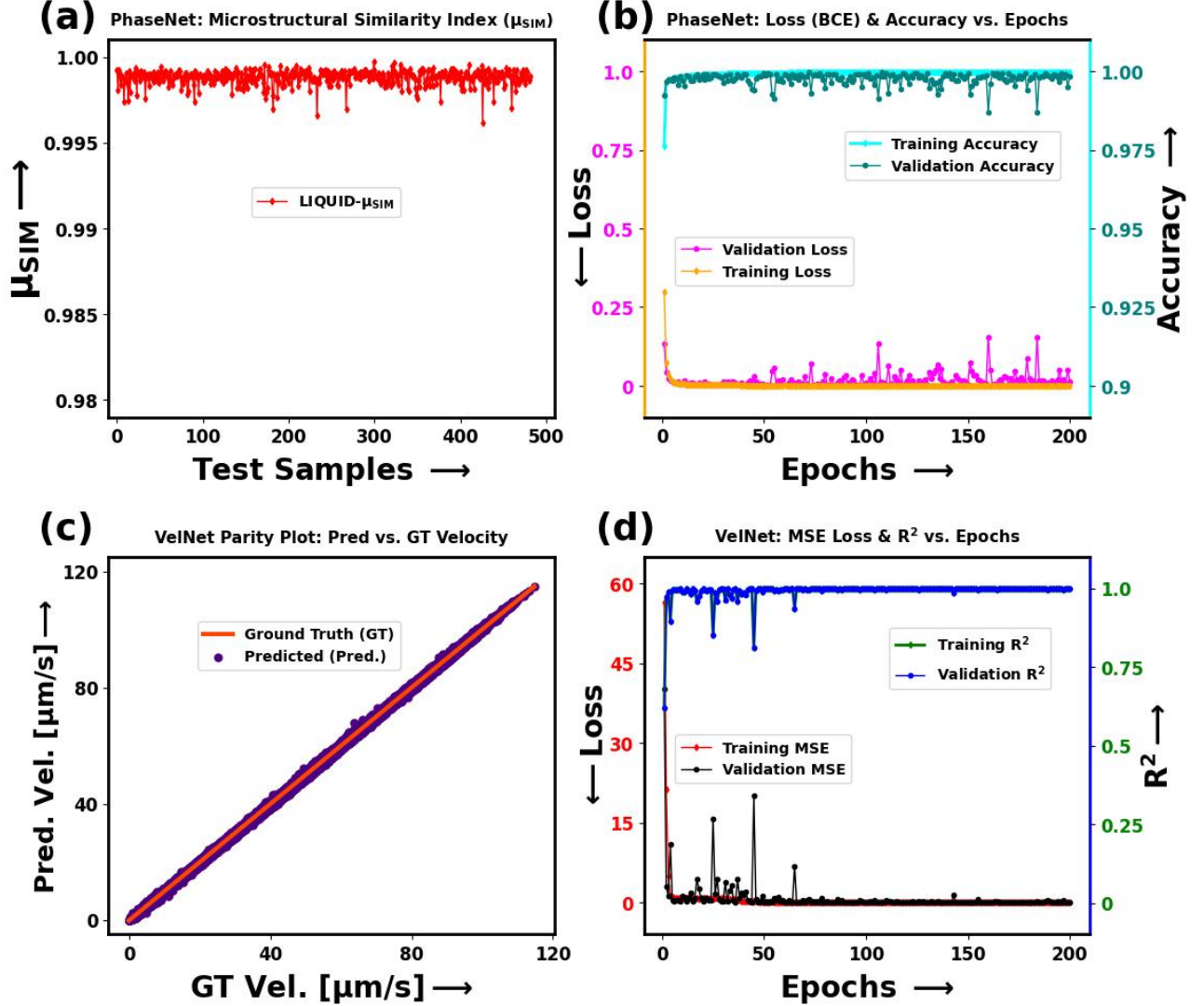


Figure 12: Training, validation, and testing performance metrics of the machine learning models. (a) Microstructural similarity index (μ_{SIM}) between the ground truth and the PhaseNet-predicted LIQUID phase morphology for the test dataset. The high average μ_{SIM} value (0.995) across 483 test samples (15% of the total dataset) confirms the model's predictive accuracy on unseen data. (b) Training and validation accuracy and loss curves for the PhaseNet model over 200 epochs. Both metrics closely follow each other, reaching high accuracy and low loss for training and validation datasets. Loss is plotted along the left y-axis, while accuracy is shown on the right y-axis. (c) Parity plot comparing VelNet-predicted velocities with ground truth velocities at each spatial point within the LIQUID phase over a $1000\mu\text{m} \times 250\mu\text{m}$ domain. The red line denotes ground truth, while purple dots represent the predicted velocities. (d) Training and validation loss (MSE) and R^2 metrics for the VelNet model over 200 epochs. Initial fluctuations stabilize after 50 iterations, showing consistently high R^2 and low loss, indicating robust training and validation performance.

are key to additive manufacturing process control and optimization. Running the architecture on a portable device enables real-time, autonomous monitoring and remote prediction without manual intervention, achieving conditional automation by dynamically adapting predictions to input conditions.

Compared to our previous work [10], where materials properties such as density, specific heat, free energy, and mobility were manually provided for FEM based phase prediction, the current approach eliminates the need for those external inputs. The trained ML models autonomously infer phase regions, areas, and melt pool velocity solely from laser-induced thermal data. Since additive manufacturing inherently involves laser-microstructure interactions,

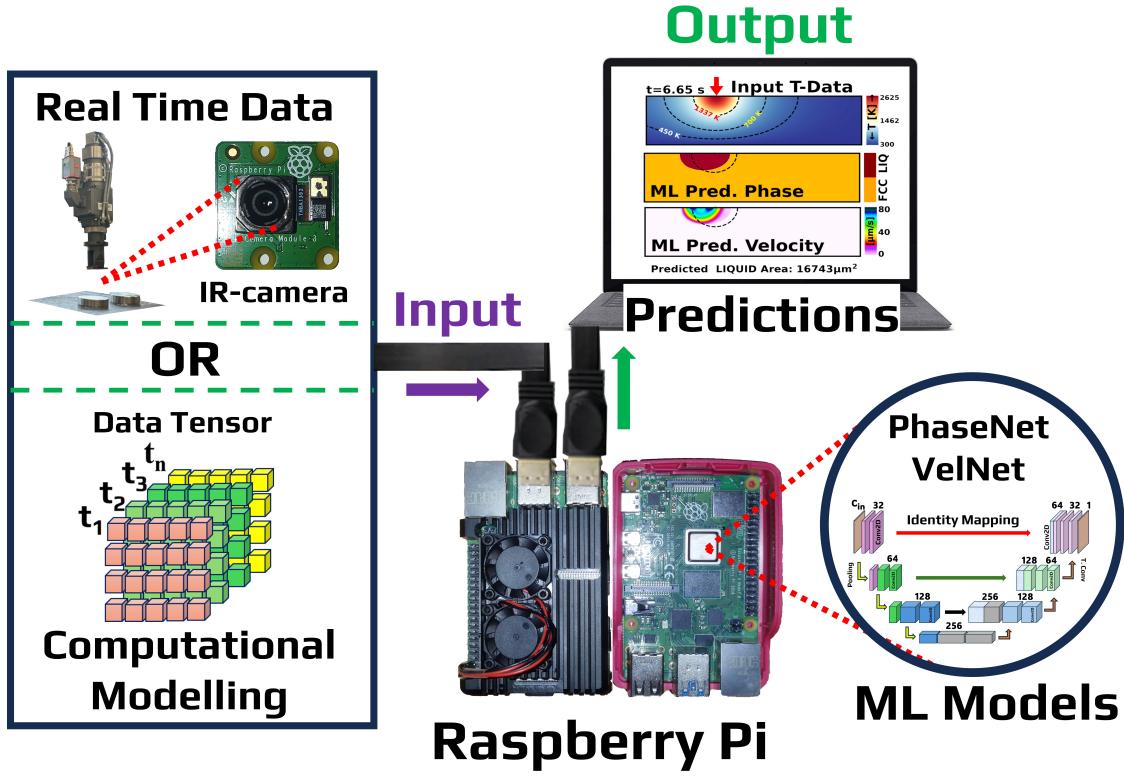


Figure 13: The I/O interface of *Raspberry Pi* micro-computer consisting the PhaseNet and VelNet programs. The input (I) interface accepts the real time image data of temperature or the data tensor of the temperature. With the help of PhaseNet model, the micro-computer in the role of MicroTwin instantly outputs the phase at the Output (O). Interestingly, the U-net architecture is reusable. The phase data from the MicroTwin can be channeled to the I interface and together with the temperature feature, it could be utilized by FlowTwin (with VelNet architecture) to predict the meltpool dynamics (velocity) in accordance with Eq. 20. The predictions of the MicroTwin and FlowTwin are elaborated in Figure 14.

this autonomy, while still relying on external laser parameters, represents a step towards Conditional Automation (L3-Automation).

10 Conclusion

This study presented computational modeling of laser-induced phase transformation and microstructural evolution in a unary Au system. Simulations were performed using four laser beam profiles: Gaussian, Flat-Top, Ring, and Bessel, combined with two different free energy formulations. A fine-tuned logarithmic-polynomial fitted expression (F_θ) and a TDB-extracted expression (F_ψ).

The Gaussian, Flat-Top, and Bessel beams exhibited similar thermal and flow characteristics, with peak temperatures and maximum meltpool velocities concentrated at the beam center. In contrast, the Ring beam produced a more uniform temperature field within its radius. A key distinction was observed between the two free energy formulations: simulation using F_θ showed a precise alignment between the melting temperature contour (1337 K) and the LIQUID-FCC interface, while those using F_ψ revealed a noticeable lead-lag behavior between the thermal and phase fronts, attributed to differing diffusion rates of heat and phase fields.

Two U-Net based machine learning models, PhaseNet and VelNet, were trained on simulations datasets linking thermal histories to phase evolution and meltpool velocity. Using these models, a virtual digital twin framework was developed, capable of predicting phase and velocity evolution from temperature data inputs. This framework demonstrates a step towards constructing a physical Digital Twin for real-time experimental prediction using infrared camera data.

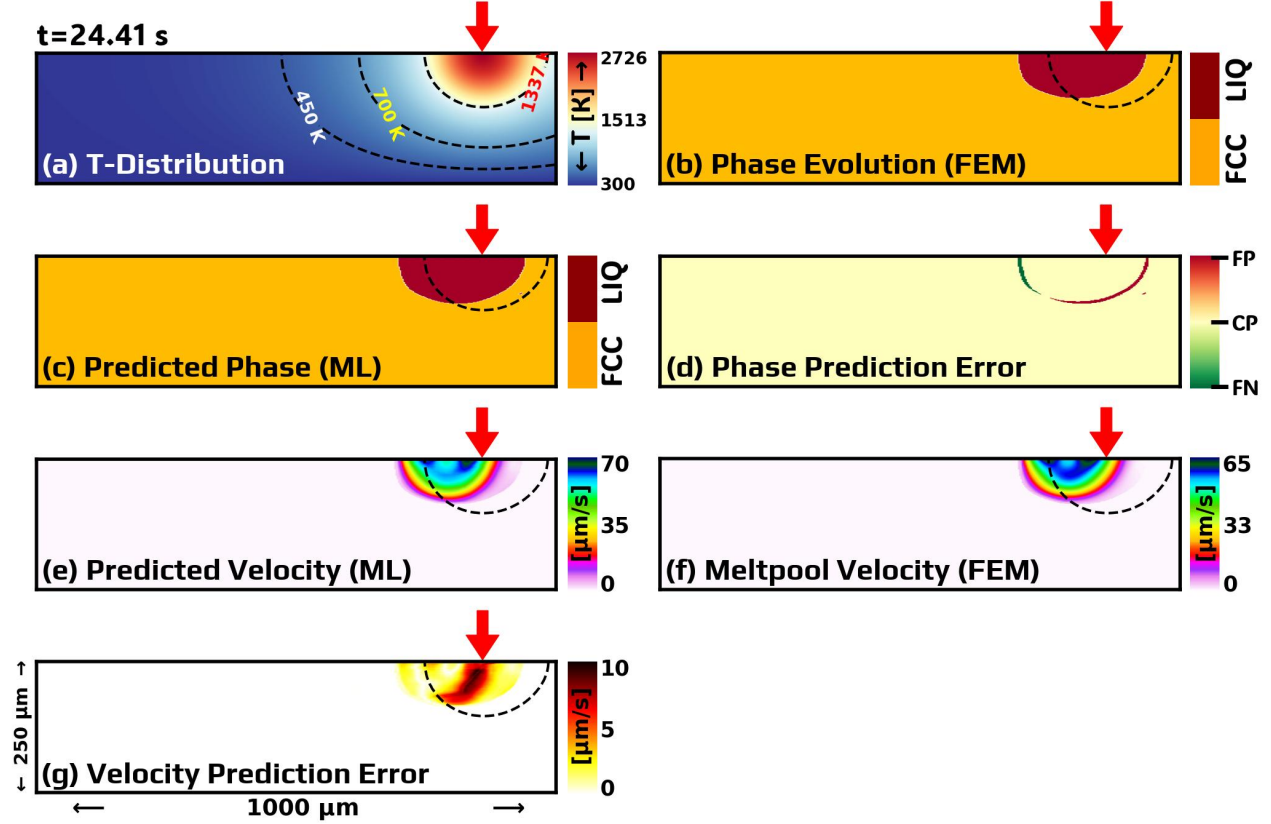


Figure 14: Comparison of phase evolution and melt pool flow velocity predicted by the PhaseNet and VelNet models with FEM-based phase-field simulations. (a) Temperature distribution used as input to both ML models, obtained from a laser heat transfer simulation with a Bessel heat source and no phase-field coupling. In a real-time digital twin setup, similar data could be acquired from an IR camera and preprocessed on a microcomputer (e.g., *Raspberry Pi*) for model input. (b) FEM-based phase-field simulation result at $t = 24.41s$ showing a LIQUID phase area of $18046 \mu m^2$. (c) MicroTwin’s PhaseNet-predicted phase distribution with a LIQUID phase area of $17363 \mu m^2$. (d) Comparison of PhaseNet predictions with FEM results, where CP (light yellow) denotes correctly predicted regions, FP (green) indicates false positives, and FN (red) marks false negatives. Most mismatches occur along the LIQUID-FCC interface. (e) FlowTwin’s VelNet-predicted flow velocity within the LIQUID phase and (f) corresponding FEM-simulated melt pool velocity for identical laser parameters. (g) Absolute velocity prediction error between VelNet and FEM results at the same time step. The input temperature field originates from a stand-alone thermal FEM simulation not included in training, validation, or testing datasets, while the FEM-based phase and velocity fields used for comparison are part of the coupled-physics simulations used during training.

At present, the model is limited to a unary Au system, but future work will extend it to multi-component alloys and a wider range of laser processing parameters. The development of such multimodal Digital Twin systems is expected to accelerate real-time phase transformation prediction and advance precision control in additive manufacturing.

11 Declarations

The authors declare to have no conflicts of interest that are relevant to the content of this article.

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13 Data Availability

The phase-field simulation codes, simulation video animations, and webapp for virtual digital twin along with the data used for this work is available at <https://github.com/subediupadesh/AMTwinLab> and the machine learning model source code is available at <https://github.com/AadyaBhattarai/PhaseVelNet>.

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Appendix A:

A1 Ramp-up rate of Driving Force ($\Lambda_{\psi|\theta}$)

To quantify the discrepancy introduced by the fitted free-energy expression, we compare the temperature derivatives of the LIQUID-FCC free energy difference over a narrow interval if $\delta T = \pm 5K$ around the melting point $T_m = 1337K$ as presented in Figure A1. This analysis highlights how the physically consistent TDB-based formulation preserves the thermodynamically consistent ramp-up rate of the phase-transition driving force (Λ), whereas the polynomial fit introduces an unrealistically steep temperature dependence. Since the magnitude of the driving force $|\Delta F|$ directly determines the predicted phase-transition kinetics, even small distortions in $\frac{d\Delta F}{dT}$ can lead to order of magnitude error in phase front, underscoring the necessity of using thermodynamically grounded free energy models.

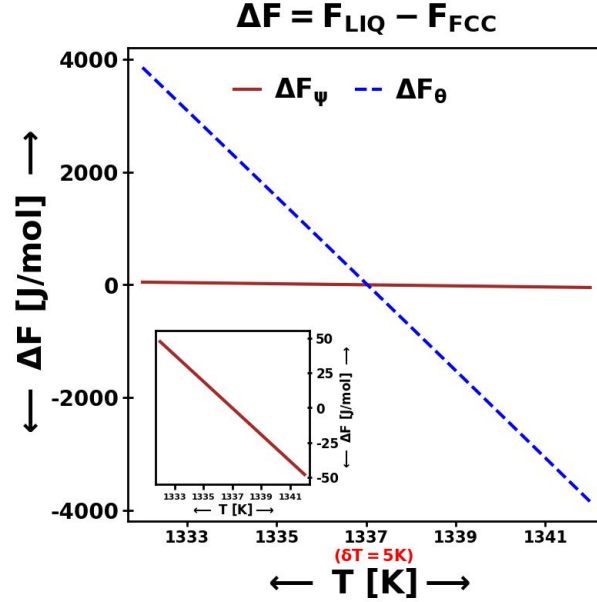


Figure A1: Comparison of the free energy ramp-up rate $\Lambda = \frac{d\Delta F}{dT}$ near melting point. The figure plots the temperature derivative of the LIQUID-FCC free energy differences, $\Delta F = F_{\text{LIQUID}} - F_{\text{FCC}}$, over a narrow interval of $\delta T = \pm 5K$ around $T_m = 1337 K$, comparing the TDB-based and fine-tuned (fitted) free energy expressions. The TDB-based formulation yields a physically consistent ramp-up rate of $\Lambda_{\psi} \approx -9.511 J/mol/K$. In contrast, the fitted expression produces an artificially steep slope of $\Lambda_{\theta} \approx -770.45 J/mol/K$, exaggerating the thermodynamic behavior by a factor of ≈ 81 . The inset highlights only the TDB-based Λ_{ψ} curve to improve visibility of its small but physically meaningful temperature dependence. This discrepancy inflates the $|\Delta F|$ and thereby the phase-transition kinetics, resulting in absence of lead-lag dynamics in simulation performed with fine-tuned free energy expression.

A2 Complementary Figures: Phase response to $F_{\psi|\theta}$

This section presents the complementary Figures and Plots A2, A3 and A4, related to Section 8.1 and 8.2.

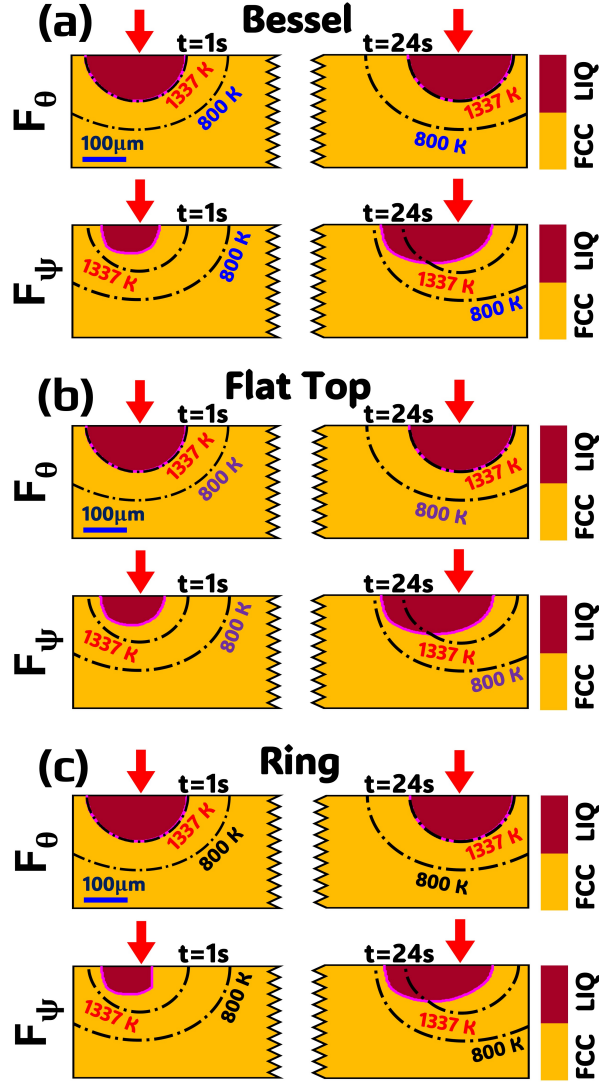


Figure A2: Visual illustration of laser induced phase evolution during the phase field simulation in unary Au system with different heat source profiles with different free energy expressions. (a) The four subplots depicts the growth of melt pool at time $t=1s$ and $t=25s$ under the Bessel beam with top row corresponding to fine-tuned free energy expression F_θ and bottom row with F_ψ . Similarly, in (b) the LIQUID phase status is presented for Flat-Top laser profile with F_θ and F_ψ as driving force for phase transformation in top and bottom row subplots. Finally (c) represents the melt pool phase growth similar to that of (a) and (b) but for the simulation with Ring laser beam under two free energy expressions. Although the different laser beam profile resulted in drastic temperature history and velocity profile inside the domain, the overall phase growth is in near similar fashion and is possible difficult to distinguish on a visual basis.

Appendix B:

B1 Microstructural Similarity Index (μ_{SIM})

The Microstructural Similarity Index, μ_{SIM} , is computed using the formula shown in Equation B1 [63, 64]. In this expression, η_t corresponds to the ground-truth phase-field values obtained from the simulation, while η'_t represents the PhaseNet-predicted values of the phase variable η at time 't'.

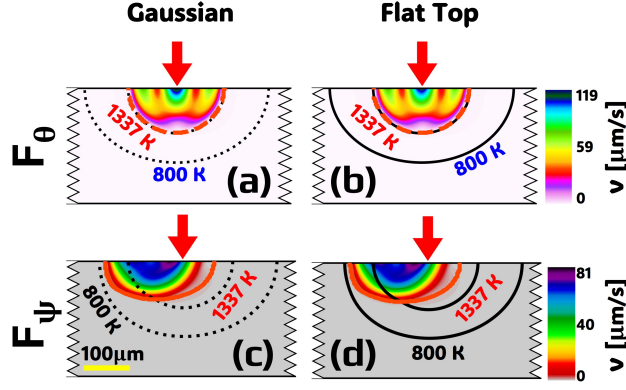


Figure A3: Illustration of velocity profile inside the meltpool when the laser traverses to domain midpoint. Subplot (a) and (c) are associated with simulation under the Gaussian heat source for F_θ and F_ψ respectively. The peak flow velocity for in the LIQUID phase for (a) and (c) are $119 \mu\text{m/s}$ and $75.61 \mu\text{m/s}$ respectively. In (b) and (d), the flow profile for simulation conducted under the Flat-Top heat source for F_θ and F_ψ is presented.

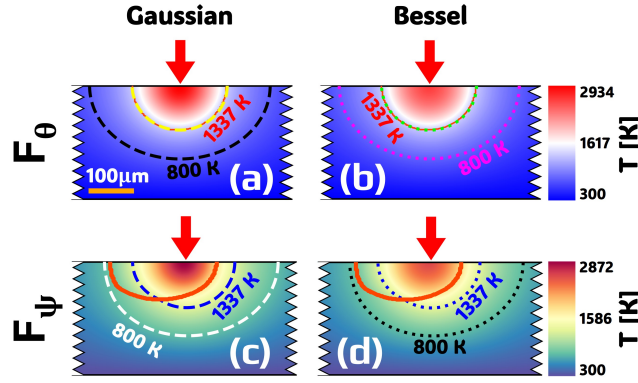


Figure A4: Visualization of thermal distribution across a cutout of simulation domain for Gaussian and Bessel laser sources. Plot (a) and (c) showcases the temperature profile for Gaussian heat source with driving force expression of F_θ and F_ψ respectively. Similarly, (b) and (d) represents the temperature inside the computational domain with Bessel heat source for F_θ and F_ψ correspondingly. A subtle difference in temperature profile can be noticed between Gaussian and Bessel based simulation results with a higher concentration of temperature distribution at the beam centre for Gaussian beam, the peak temperature reached by the domain under Bessel beam are found to be 2613.11 K in (b) and 2548.95 K in (d).

$$\mu_{SIM}(\eta_t, \eta'_t) = \frac{(2\mu_{\eta_t}\mu_{\eta'_t} + \alpha_{\eta_t})(2\sigma_{\eta_t\eta'_t} + \alpha_{\eta'_t})}{(\mu_{\eta_t}^2 + \mu_{\eta'_t}^2 + \alpha_{\eta_t})(\mu_{\eta_t}^2 + \mu_{\eta'_t}^2 + \alpha_{\eta'_t})} \quad (\text{B1})$$

The quantities μ_{η_t} , $\mu_{\eta'_t}$, $\sigma_{\eta_t}^2$, and $\sigma_{\eta'_t}^2$ denote the sample mean intensities and variances of the fields η_t and η'_t , respectively, while $\sigma_{\eta_t\eta'_t}$ represents their cross-covariance. The constants α_{η_t} and $\alpha_{\eta'_t}$ are included to avoid division by zero. For the μ_{SIM} metric to be computed correctly, both ground truth distribution of η_t and predicted η'_t distribution must share the same data points in 2D domain of $N \times M$.

Appendix C:

C1 Virtual Digital Twin Work Flow

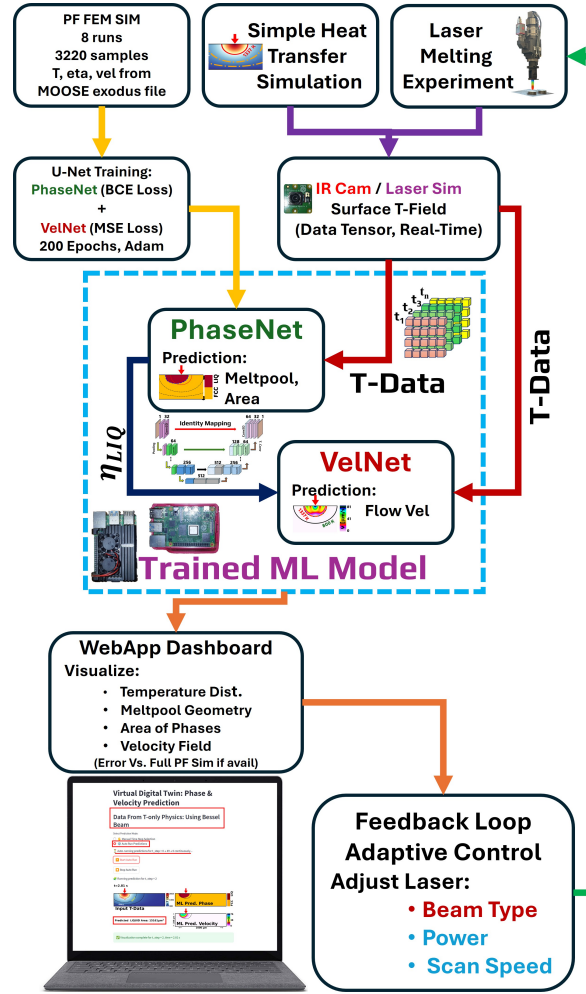


Figure C1: Flowchart of the Virtual Digital Twin framework for real-time melt-pool prediction and laser control. High-fidelity Phase-Field (PF) simulations (8 runs, 3220 samples) provide temperature, phase, and velocity fields used to train two U-Net models: PhaseNet for melt-pool geometry and phase distribution (BCE loss) and VelNet for flow-velocity prediction (MSE loss). Independent temperature-field inputs, either from a simple heat-transfer laser simulation or real-time IR-camera measurements during laser melting, can be supplied to the trained models to infer melt-pool shape, phase area, and velocity fields. These predictions are displayed in a WebApp interface for visualization and optional comparison to PF simulations. The interface enables a feedback-loop adaptive control system that adjusts laser processing parameters (beam type, power, scan speed), closing the Virtual Digital Twin for real-time monitoring and control.

Appendix D:

D1 Computational Cost Comparison

The phase-field simulations were executed on a *Dell Precision 7920 Tower* using 10 CPU cores (2 threads each), 64 GB RAM, and an Intel Xeon Gold 5220R @ 2.20 GHz, launched via *mpiexec* within our MOOSE-based application. The average wall-clock time for a single FEM simulation was ≈ 12.5 hours, and the eight simulations required for this study accumulated to roughly 4.5 days of compute time. In contrast, training the ML models (**PhaseNet** and **VelNet**) for 200 iterations on an *NVIDIA A100 GPU* required only 30-45 minutes. The largest performance gain arises during inference, where the trained ML models generate full spatiotemporal fields for new thermal inputs in **under 1 second**, representing a speed-up of five orders of magnitude relative to FEM simulations. This behavior was validated across multiple platforms, including a *Raspberry Pi 4*, a high-performance workstation, and a freely hosted Streamlit web application, all achieving sub-second inference. Table D1 summarizes these inference times.

Table D1: Inference time using trained ML model across different systems

	Inference Time
Raspberry Pi 4	≈ 0.265 s
Dell Precision 7920 Tower	≈ 0.192 s
Cloud Computing via Streamlit	≈ 0.397 s

These results demonstrate that the ML models enable real-time prediction of phase morphology, phase-fraction evolution, and melt pool flow velocity, in contrast to the hours-long FEM computations required for each new condition. This real-time capability is also integrated into the Virtual Digital Twin (VDT) interface, where inference time per prediction is displayed. Overall, the ML-based workflow provides a robust and quantitatively verified acceleration over FEM simulations, supporting its suitability for real-time process control and monitoring.

D2 Convergence Behavior Across Diffused-Interface Widths

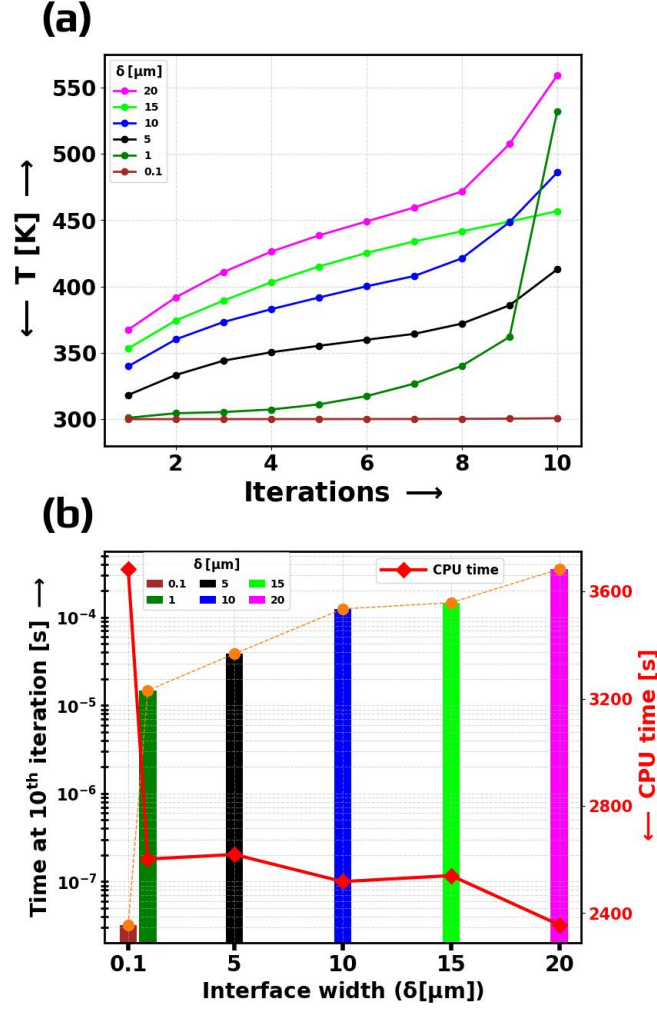


Figure D1: Effect of diffused-interface width δ on convergence behavior and computational cost. The same Gaussian TDB-based phase-field simulation was performed for $\delta = 0.1, 1, 5, 10, 15$ and $20 \mu\text{m}$, each run for 10 iterations. Subplot (a) shows the maximum temperature in the domain at the 10th iteration, demonstrating the simulations with larger diffused-interface widths reach higher temperatures more rapidly for the same iteration count. Subplot (b) explains this behavior via the actual physical simulation time (left y-axis) achieved after 10 iterations. For the smallest interface width ($\delta = 0.1 \mu\text{m}$), the simulation advances only to $\sim 10^{-8}\text{s}$, because the sharp interface causes repeated convergence failure; the simulation therefore is programmed to reduce the timestep dt until convergence is possible, significantly increasing the wall clock time (right y-axis), which exceeds 3600s for 10 iterations. In contrast, larger δ values permit larger stable timesteps, leading to higher physical simulation times and substantially lower CPU times. Based on this balance between accuracy, convergence stability and computational efficiency, a diffused-interface width of $\delta = 10 \mu\text{m}$ was selected for the simulation in this work.